

Dancing on the Saddles: A Geometric Framework for Stochastic Equilibrium Dynamics[†]

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Abstract

I introduce *conditional saddle paths*: equilibrium trajectories under frozen exogenous regimes, traced by stochastic-equilibrium decision rules. Their *geometry* diagnoses state dependence: shock responses depend on position along the saddle and on slope gaps across regimes. Their *monotonicity* delivers dimension reduction: within each frozen-regime spell, aggregate capital exactly pins down the wealth distribution in [Krusell and Smith \(1998\)](#), under explicit conditions verified in the standard calibration. A synchronization argument explains why capital-alone forecasting remains nearly exact globally. In a search-and-matching economy, unemployment volatility is hump-shaped in the benefit level: at high benefits, the steady states under good and bad productivity converge, leaving little room for unemployment to fluctuate between them.

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1 Introduction

In deterministic dynamic models, a single saddle path traces the economy’s transition to steady state, and a phase diagram immediately conveys the qualitative dynamics. No comparable geometric framework exists for stochastic environments. Two obstacles stand in the way. First, stochastic switching creates *multiple* equilibrium loci—one per exogenous state—so no single saddle path exists. Second, heterogeneous-agent models have an infinite-dimensional wealth distribution as their endogenous state. Geometric visualization is infeasible, and dimension reduction becomes empirical rather than structural. This paper overcomes both obstacles by introducing a pair of linked geometric objects: frozen-regime conditional saddle paths and their regime-specific policy graphs. Policy-graph slopes diagnose state dependence; monotonicity along frozen-regime paths delivers dimension reduction.

The paper makes three contributions. First, I define conditional saddle paths as equilibrium trajectories under frozen exogenous regimes, computed under the decision rules of the stochastic equilibrium, and distinguish them from the policy graphs traced by those trajectories in aggregate phase diagrams. These objects decompose business-cycle fluctuations into movements *along* a conditional saddle (endogenous propagation) and *across* conditional saddles (exogenous transitions), providing a stochastic analogue of deterministic phase diagrams. State-dependent shock responses have a precise geometric signature: impact gaps are state-independent exactly when the relevant policy graphs are vertical translations of one another, and slope differences are the local first-order diagnostic of deviations from that benchmark. Even when impact responses are nearly state-independent, the *shape* of the impulse response—peak timing and persistence—can be highly state-dependent, driven by distance along the saddle rather than slope asymmetry. I develop these results in a canonical RBC model (Sections 2–3).

Second, I provide a dimension-reduction theorem. If an aggregate variable is strictly monotone along each conditional saddle, it is a *conditional sufficient statistic*: it exactly indexes all equilibrium allocations and prices within each frozen-regime spell. I prove that aggregate capital serves this role in [Krusell and Smith \(1998\)](#). Near the conditional steady state, the result needs only standard linearization regularity. Away from it, it holds on an explicit region under bounds on household consumption responses—inequalities that can be checked from calibrated parameters—with one residual condition, ruling out states where aggregate capital stalls while the distribution keeps moving, stated as an explicit hypothesis of the theorem. An illustration with non-concave production confirms that the underlying geometric condition is not automatic: budget-feasible transitions violating it exist once concavity is dropped. Different histories can leave different distributions behind the same

capital stock. A synchronization argument bounds this residual variation: when the frozen-regime dynamics contract, the current state is determined by the recent regime history alone, so two economies whose regimes have recently agreed must be close, however different their distant pasts. Near the conditional steady state, the speed at which the past is forgotten is the same rate that governs convergence along the saddle. I also develop a perfect-foresight diagnostic that verifies the key monotonicity condition without solving the full stochastic model (Section 4).

Third, I apply the framework to a search-and-matching economy (Section 5), where the saddle geometry organizes the positive analysis of how policy parameters reshape equilibrium volatility. Conditional saddles trace upward-sloping, regime-specific vacancy loci whose across-regime transitions generate the empirical Beveridge-curve loop, and the surplus compression behind the Shimer puzzle manifests as differential saddle steepness. Comparative statics on unemployment benefits yield a new result: volatility is hump-shaped in the benefit level. The saddle geometry diagnoses why. What governs volatility is the *separation between conditional steady states*, not slope asymmetry alone. At high benefit levels, conditional steady states converge despite widening slope gaps, leaving little room for unemployment to fluctuate between regimes.

Related literature This paper contributes to three strands of literature in macroeconomics. The first is the literature studying global equilibrium dynamics and solution methods under aggregate uncertainty. The challenge of characterizing equilibrium dynamics in models with aggregate shocks has motivated extensive methodological development. [Marcet \(1988\)](#), [Den Haan and Marcet \(1990\)](#), and [Krusell and Smith \(1998\)](#) pioneered the use of bounded rationality approximations. In particular, [Krusell and Smith \(1998\)](#) discovered that a simple linear forecasting rule in aggregate capital achieves remarkable accuracy ($R^2 > 0.9999$) despite the infinite-dimensional state space. Subsequent work has sharpened accuracy and efficiency through diverse approaches ([Den Haan, 1996](#); [Reiter, 2010](#); [Auclert et al., 2021](#); [Fernández-Villaverde et al., 2023](#), among others).

The sequence-space Jacobian (SSJ) method of [Auclert et al. \(2021\)](#) provides a complementary perspective on dimension reduction. SSJ linearizes around a steady state; dimension reduction follows from the additive structure of linearized IRFs. By contrast, conditional saddle paths operate globally and nonlinearly: dimension reduction follows from monotonicity of an aggregate coordinate along the frozen-regime equilibrium path. Near a conditional steady state, the policy graph’s tangent connects to the SSJ representation. Away from it, the conditional saddle captures nonlinear dynamics—including state-dependent response shapes (Section 3)—outside the scope of first-order perturbation.

This paper differs from these computational contributions by providing geometric foundations for *why* dimension reduction works. Aggregate capital’s within-spell sufficiency follows from geometric properties—nullcline invariance and monotonicity—rather than numerical happenstance. The result also has computational implications: along each conditional saddle, aggregate capital provides a theoretically justified metric for the repeated transition method of Lee (2025).¹

Second, this paper extends the geometric tradition of phase-diagram analyses in growth theory (Solow, 1956; Swan, 1956; Ramsey, 1928; Koopmans, 1963; Cass, 1965; Barro and Sala-i Martin, 1992) to stochastic environments. Conditional saddle paths provide a visual tool for understanding how microfounded frictions generate state-dependent dynamics, as documented empirically (Kaplan and Violante, 2014; Vavra, 2014; Berger and Vavra, 2015; Basu and Bundick, 2017; Bloom et al., 2018; Kaplan et al., 2018; Petrosky-Nadeau et al., 2018; Baley and Blanco, 2019; Pizzinelli et al., 2020; Berger et al., 2021; Winberry, 2021; Lee, 2026; Melcangi, 2024).

Finally, in the random dynamical systems literature (Arnold, 1998; Schenk-Hoppé, 1998; Schenk-Hoppé, 2001), the focus is on history-dependent stochastic saddle paths (Yannacopoulos, 2011), which vary with the realized shock sequence.² By contrast, conditional saddle paths are history-invariant geometric objects that provide economists with the direct analogue of deterministic phase diagrams (Kamihigashi, 2003, 2005).

2 Conditional saddle path

2.1 Definitions and assumptions

I consider a generic dynamic stochastic model where the corresponding recursive competitive equilibrium (RCE) is characterized by the following aggregate state S and the endogenous

¹The idea of solving heterogeneous-agent models by freezing the aggregate state and analyzing transitions appears in several computational approaches, including Bourany (2018), Section 4.3 of a working paper version of Achdou et al. (2021), and Lee (2025). In continuous time, the master-equation literature (Achdou et al., 2021; Bilal, 2023) treats the distribution itself as the state and asks when finite-dimensional projections suffice—the rigorous mathematical home of the question studied here. The present paper provides a theoretical foundation: conditional saddle paths formalize the underlying geometric objects, and the sufficiency results explain when dimension reduction is exact along frozen-regime paths.

²In Random Dynamical Systems language, the conditional steady state is a deterministic realization of the random fixed point (Schenk-Hoppé and Schmalfuß, 2001), and the conditional saddle path is the invariant manifold of the frozen-regime dynamics. The connection is put to work in Section 4: the synchronization theorem (Theorem 2), which underlies the across-spell bound (Proposition 9), uses exactly the pullback random-fixed-point construction of Schenk-Hoppé and Schmalfuß (2001).

and exogenous laws of motion $(\Gamma_{\text{endo}}, \Gamma_{\text{exo}})$:

$$S = [\Phi, A] \tag{1}$$

where Φ is the endogenous aggregate state variable, and A is the exogenous aggregate state variable. The latter may be a multivariate vector that follows a stochastic process. I assume the exogenous aggregate law of motion Γ_{exo} is a Markov chain. For simplicity in the illustration, I assume Γ_{exo} is a two-state Markov chain where $A \in \{B, G\}$, and $\Gamma_{\text{exo}}(A'|A) > 0 \forall A', A$.

I consider a distributional state space $\mathcal{X}$, whose elements $\Phi \in \mathcal{X}$ summarize the cross-sectional distribution of idiosyncratic household states that are payoff-relevant for equilibrium (e.g., assets and employment/productivity types) together with any endogenous objects needed to evaluate equilibrium decision rules. Formally, one can take $\mathcal{X}$ to be a subset of a metric space of probability measures augmented, if needed, by a finite-dimensional vector of aggregate variables. Throughout, $\Gamma_{\text{endo}}(\cdot, A) : \mathcal{X} \rightarrow \mathcal{X}$ denotes the recursive equilibrium law of motion mapping the current distributional state into the next-period state under frozen regime A .

The central object of this paper formalizes a natural thought experiment. Suppose the economy is in state (Φ_0, A) today and the exogenous regime happens to remain at A indefinitely. The resulting equilibrium trajectory—computed under decision rules that fully internalize future regime changes—is a frozen-regime orbit. Its closure is the conditional saddle path. In aggregate phase diagrams, the corresponding regime-specific locus is a conditional policy graph; slope and curvature statements below refer to these graph projections.

Definition 1 (Frozen-regime orbit and conditional saddle path).

Fix a regime $A \in \{B, G\}$ and an initial state $\Phi_0 \in \mathcal{X}$. Let $\{\Phi_t\}_{t \geq 0}$ denote the frozen-regime continuation under A , defined recursively by

$$\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A), \quad t \geq 0. \tag{2}$$

The frozen-regime orbit is

$$\mathcal{O}(\Phi_0, A) := \{\Phi_t : t \geq 0\}. \tag{3}$$

The conditional saddle path under A from Φ_0 is defined as the closure of the frozen-regime orbit:

$$\mathcal{M}(\Phi_0, A) := \overline{\mathcal{O}(\Phi_0, A)}. \tag{4}$$

Throughout, the distributional state space is a metric space of probability measures $(\mathcal{X}, d)$, where d is compatible with weak convergence of distributions. The closure in Definition 1 is taken with respect to this metric.

Two projection ingredients recur throughout: a *coordinate* $e : \mathcal{X} \rightarrow \mathbb{R}$, a predetermined aggregate variable such as aggregate capital K or inherited unemployment u_{-1} ; and an *aggregate object* $g(\Phi, A)$, an equilibrium outcome such as consumption or vacancies.

Definition 2 (Conditional policy graph).

For a regime A , a coordinate e , and an aggregate object g , the conditional policy graph in the (e, g) projection is

$$\mathcal{P}_A^g := \{(e(\Phi), g(\Phi, A)) : \Phi \in \mathcal{D}_A\}, \quad (5)$$

where $\mathcal{D}_A$ is the set of distributional states reached by the frozen- A continuations under study.

When e is injective on a given conditional saddle path, the restriction of $\mathcal{P}_A^g$ to that path is the graph of a single-valued function of e .

Based on the conditional saddle path, I define the conditional steady state as follows:

Definition 3 (Conditional steady state).

Given (Φ_0, A) , if the frozen-regime continuation $\{\Phi_t\}_{t \geq 0}$ converges, I denote its limit by

$$\Phi^{cs}(\Phi_0, A) := \lim_{t \rightarrow \infty} \Phi_t. \quad (6)$$

Definition 1 defines the conditional saddle path directly from the frozen-regime continuation. Fixing A makes the equilibrium law of motion deterministic on the distribution space: starting from Φ_0 , the sequence $\{\Phi_t\}_{t \geq 0}$ is generated by repeated application of the endogenous transition operator $\Gamma_{\text{endo}}(\cdot, A)$. In this sense, $\mathcal{M}(\Phi_0, A)$ is the natural analogue of the saddle arm in deterministic saddle-path analysis. It isolates the equilibrium states visited (and their limit points) along the convergence dynamics under regime A .³

For the conditional saddle to serve as a useful geometric object—a one-dimensional curve that the economy traverses during each frozen-regime spell—the frozen-regime dynamics must converge to a unique limit and vary smoothly. The following assumption collects these regularity conditions.

³Strictly speaking, $\mathcal{M}(\Phi_0, A) = \overline{\mathcal{O}(\Phi_0, A)}$ in discrete time is an orbit-closure, not necessarily a smooth manifold. I use saddle-path language to emphasize its one-dimensional, path-like role. Slope and curvature language applies to the conditional policy graph $\mathcal{P}_A^g$ in a chosen aggregate projection; in continuous time, the corresponding frozen-regime object is a one-dimensional invariant manifold; see Appendix L for the full continuous-time formulation.

Assumption 1 (Regularity of conditional saddle paths).

Fix a regime $A \in \{B, G\}$ and an initial state $\Phi_0 \in \mathcal{X}$. Let $\{\Phi_t\}_{t \geq 0}$ be defined by $\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A)$.

- (i) (Unique existence) The conditional steady state $\Phi^{cs}(\Phi_0, A)$ uniquely exists on $\mathcal{M}(\Phi_0, A)$.
- (ii) (Continuity) The map $\Gamma_{\text{endo}}(\cdot, A) : \mathcal{X} \rightarrow \mathcal{X}$ is continuous. Moreover, the aggregate variables (e.g., aggregate capital K and consumption C) vary continuously in Φ when restricted to $\mathcal{M}(\Phi_0, A)$.

Assumption 1 collects the regularity properties needed for the paper’s geometric arguments. Part (i) requires that, conditional on a regime A and an initial state Φ_0 , the frozen-regime continuation converges to a unique conditional steady state. This does not assert global uniqueness across all initial conditions, nor does it rule out multiplicity across regimes.⁴ Part (ii) imposes continuity of the frozen-regime law of motion and aggregate observables on the conditional saddle, ensuring that small perturbations of the distributional state produce small changes in aggregate outcomes. Assumption 1(i) thus restricts attention to convergent frozen-regime continuations, ruling out periodic orbits or divergent paths. In the models studied below, convergence follows from standard arguments: diminishing returns in the RBC benchmark and contractivity of the Bewley–Huggett–Aiyagari operator (Bewley, 1986; Huggett, 1993; Aiyagari, 1994) in the heterogeneous-household economy.

With this regularity, the conditional saddle admits a natural notion of position along the frozen-regime orbit. Theorem 1 (Section 4) formalizes the key implication: if some aggregate equilibrium variable moves strictly one-way along this path, it provides a valid coordinate for the entire conditional saddle, yielding an exact one-dimensional representation on $\mathcal{M}(\Phi_0, A)$.

2.2 Representative-agent economy: A canonical RBC

For illustrative purposes, I consider a representative household with temporal log utility. Given initial condition (a_0, A_0) , the household maximizes lifetime utility under stochastic aggregate TFP A_t . I present the economy in recursive form and work with a recursive competitive equilibrium. Under standard regularity conditions, this formulation is equivalent to the sequential equilibrium; the sequential formulation is in Appendix A. The recursive

⁴In particular, Proehl (2025) establishes global existence and uniqueness under parametric restrictions, whereas Assumption 1 (i) allows for global multiplicity as in Walsh and Young (2024). Nevertheless, conditional on a given initial state and a fixed regime, the equilibrium path is assumed to be uniquely determined.

form is as follows:

$$v(a; K, A) = \max_{c, a'} \log(c) + \beta \mathbb{E}v(a'; K', A') \quad (7)$$

$$c + a' = a(1 + r(K, A)) + w(K, A) \quad (8)$$

$$a' \geq -\bar{a} \quad (9)$$

$$K' = \Gamma_{\text{endo}}(K, A), \quad A' \sim \Gamma_{\text{exo}}(A'|A), \quad (10)$$

where v is the household's value function; a is wealth at the beginning of a period; K is aggregate capital; A is aggregate TFP; Γ_{endo} is the law of motion for K . For illustrative purposes, I assume that TFP A follows a Markov-switching process between the levels B and G , where $G > B$:

$$\Gamma_{\text{exo}} = \begin{bmatrix} \pi_{BB} & \pi_{BG} \\ \pi_{GB} & \pi_{GG} \end{bmatrix} \quad (11)$$

I consider competitive factor prices given a CRS Cobb–Douglas production function:

$$r(K, A) = A\alpha K^{\alpha-1} - \delta \quad (12)$$

$$w(K, A) = A(1 - \alpha)K^{\alpha}, \quad (13)$$

where K is capital stock, which satisfies $K = a$ in equilibrium. The recursive competitive equilibrium (RCE, hereafter) is as follows:

Definition 4 (Recursive competitive equilibrium).

$(c, a', v, r, w, \Gamma_{\text{endo}})$ is a recursive competitive equilibrium if:

1. given prices (r, w) , the policies (c, a') attain the value v in the household problem;
2. prices satisfy the competitive factor-price conditions above;
3. consistency holds:

$$a'(K; K, A) = K' = \Gamma_{\text{endo}}(K, A). \quad (14)$$

Based on the endogenous law of motion Γ_{endo} defined in the recursive competitive equilibrium of Definition 4, I define the conditional K -nullcline.

Definition 5 (Conditional K -nullcline in RBC).

Fix a regime $A \in \{B, G\}$. The conditional K -nullcline is the zero-net-investment locus in

the (K, C) phase space,

$$\mathcal{N}(A) := \{(K, C) \in \mathbb{R}_{++} \times \mathbb{R}_{++} : \Delta K(K, C; A) = 0\}, \quad (15)$$

where $\Delta K(K, C; A)$ is the one-period change in capital implied by the resource constraint at (K, C) under regime A . When $\mathcal{N}(A)$ is the graph of a function, I write $\mathcal{N}(A) = \{(K, C) : C = C_A^{Knull}(K)\}$.

The locus $\mathcal{N}(A)$ is distinct from the *realized nullcline crossings* along a frozen-regime continuation, $\{(K_t, C_t) : t \geq 0, K_{t+1} = K_t\}$, which for a strictly monotone continuation reduce to the conditional steady-state point alone. In the RBC model, the aggregate resource constraint $Y = C + I$ with $I = K_{t+1} - (1 - \delta)K_t$ gives $\Delta K = AK^\alpha - \delta K - C$, so $\mathcal{N}(A)$ is the graph of an explicit function.

Proposition 1 (Characterizing the conditional K -nullcline in RBC).

The conditional nullclines of aggregate capital K for $A \in \{B, G\}$ are

$$C_A^{Knull}(K) = AK^\alpha - \delta K, \quad (16)$$

independent of any initial condition.

Proof. From $\Delta K(K, C; A) = 0$ and $\Delta K = AK^\alpha - \delta K - C$, the nullcline is $C = AK^\alpha - \delta K$ for each A . ■

In discrete time, capital can jump by a finite amount in one period, so the projected (K, C) trajectory can develop folds or loops. These pathologies are absent in continuous time, where capital moves continuously and cannot skip past the $\dot{K} = 0$ locus. Appendix L develops the continuous-time formulation and shows that monotonicity holds without any segment-crossing restriction. In the discrete-time setting, I rule out these pathologies with the following geometric restriction.

Assumption 2 (No segment crossing of the conditional K -nullcline).

For every pre-limit transition—every $t \geq 0$ with $K_{t+1} \neq K_t$ —the open straight line segment in (K, C) connecting (K_t, C_t) and (K_{t+1}, C_{t+1}) does not intersect the conditional K -nullcline, for all $A \in \{B, G\}$:

$$((K_t, C_t), (K_{t+1}, C_{t+1})) \cap \mathcal{N}(A) = \emptyset, \quad \forall t \geq 0 \text{ with } K_{t+1} \neq K_t. \quad (17)$$

Endpoints are not required to lie off the nullcline; in particular, the conditional steady state is allowed to lie on $\mathcal{N}(A)$, and the assumption imposes no condition at the degenerate terminal transition $K_{t+1} = K_t$.

Assumption 2 ensures consecutive points remain on the same side of the K -nullcline: the nullcline gap $H_t := C_t - C_A^{Knull}(K_t)$ satisfies $H_t H_{t+1} > 0$ for every transition t with $K_{t+1} \neq K_t$. Net investment therefore cannot reverse sign along nondegenerate transitions. Combined with the no-nonterminal-stationarity condition— $K_{t+1} = K_t$ only at $\Phi_t = \Phi_A^{cs}(\Phi_0, A)$, which is guaranteed by Assumption 1(i) together with unique K -stationarity along the continuation—this yields strict K -monotonicity up to convergence. Economically, Assumption 2 rules out single-period overshooting so dramatic that capital jumps past its stationary level and reverses direction—the discrete-time analogue of the continuous-time property that trajectories cannot cross nullclines.⁵

Although Assumption 2 is stated here for the RBC benchmark, it plays a central role in the heterogeneous-household economy (Section 4). The assumption can be verified from model primitives without solving the full stochastic equilibrium, conditional on a simulated frozen-regime continuation (or certified envelope bounds on the relevant moments along it); Sections 4.1 and 4.2 develop two complementary strategies.

Proposition 2 (K monotonicity in RBC).

Fix $A \in \{B, G\}$. Under Assumptions 1 and 2, together with unique K -stationarity— $K_{t+1} = K_t$ only at $\Phi_t = \Phi_A^{cs}(\Phi_0, A)$ along the frozen- A continuation—aggregate capital converges to K_A^{cs} strictly monotonically.

Proof. Fix A and let $H_t := C_t - C_A^{Knull}(K_t)$ denote the nullcline gap. The resource constraint gives $\Delta K_t = -H_t$. By Assumption 2, $H_t H_{t+1} > 0$ for every transition t with $K_{t+1} \neq K_t$, so $\{\Delta K_t\}$ has constant nonzero sign along nondegenerate transitions. Unique K -stationarity rules out nonterminal flat steps ($\Delta K_t = 0$ at any $\Phi_t \neq \Phi_A^{cs}$), so every pre-convergence transition is nondegenerate and $\{K_t\}$ is strictly monotone. Convergence $K_t \rightarrow K_A^{cs}$ follows from Assumption 1. ■

Unique K -stationarity as a no-plateau condition Economically, unique K -stationarity rules out *plateau equilibria*: distributional states off the conditional steady state at which aggregate capital stops moving ($\Delta K_t = 0$) while the underlying distribution continues to evolve. It excludes the kind of degenerate fixed points of K that arise in models with poverty traps, occasionally binding constraints that fully bind over a range, or policy kinks that neutralize net investment at non-steady-state capital levels. In the representative-agent RBC, K is Markov under the frozen regime, so unique K -stationarity reduces to uniqueness of the conditional steady state (Assumption 1(i)). In the heterogeneous-agent economy, it holds

⁵For related geometric restrictions in dynamic optimization, see Reddy et al. (2020) and, on threshold (Skiba) phenomena, Skiba (1978), Dechert and Nishimura (1983), and Wagener (2003).

locally via the stable eigenvalue $\lambda_A^s \in (0, 1)$. A globally certified version is more delicate: the bounded price-income MPC of Assumption 3 (Appendix G) constrains the dynamics off the conditional steady state, but I treat unique K -stationarity as a maintained pathwise condition—directly checkable along any simulated frozen-regime continuation by inspecting whether $\Delta K_t = 0$ occurs only at $K = K_A^{cs}$.

Figure 1 depicts the conditional saddle paths of the calibrated RBC model in the (K, C) plane under $A \in \{B, G\}$, solved globally via the repeated transition method of Lee (2025).⁶ Panel (a) zooms into the conditional steady states and overlays a histogram of simulated (K_t, C_t) outcomes; panel (b) shows the global partition with conditional K -nullclines and direction arrows. In the long run, all equilibrium dynamics are confined to the region bounded by the two conditional saddles: within each spell, the economy converges along the active saddle, and regime switches move it across, so the entire ergodic distribution is nested within the saddle boundaries.

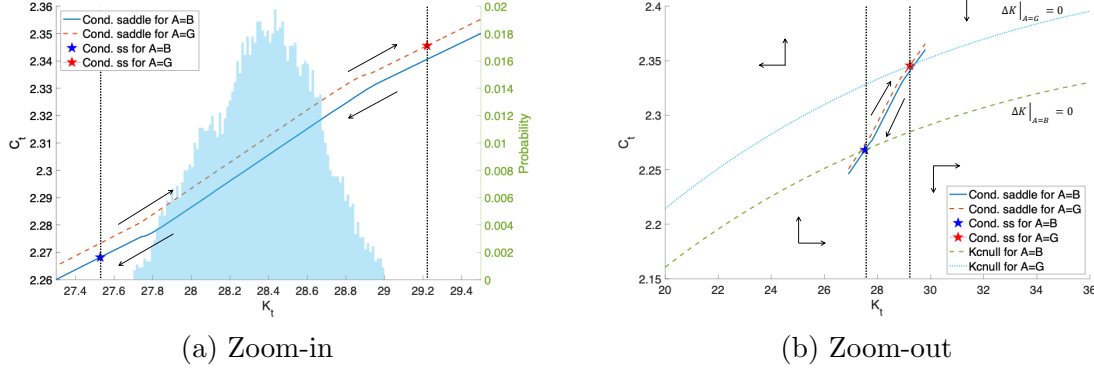
MIT shock on the saddle This decomposition yields a sharp characterization of unexpected transitory shocks (MIT shocks), whose connection to first-order perturbation in heterogeneous-agent models is established by Boppart et al. (2018). I restrict attention here to shocks representable as displacements of the predetermined state under unchanged equilibrium decision rules. For this class of shocks, there exists a shift along the conditional saddle path that replicates the shock’s impact, and all subsequent dynamics are along-path convergence. Consequently, the slope and curvature of the conditional policy graph at the point of displacement fully determine the economy’s propagation of the shock, including its persistence, amplification, and state dependence.

Generalized transition function The conditional saddle paths capture all equilibrium transitions—along saddles (endogenous propagation) and across them (exogenous regime shifts)—enabling generalized transition function (GTF) analysis (Lee, 2025), which encompasses generalized impulse response functions (Koop et al., 1996; Andreasen et al., 2017). The economy’s response to a sequence of exogenous shocks is not confined to local dynamics around a steady state: its trajectory is sharply traced in the phase diagram as a sequence of along-saddle and across-saddle movements.

Figure 2 illustrates the equilibrium dynamics in the phase diagram (panel (a)) and in the time domain (panels (b) and (c)) for a 20-quarter subsample. Negative TFP shocks produce downward jumps in consumption across conditional saddle paths, followed by endogenous

⁶For each regime A , I hold productivity fixed for 2,500 periods and iterate the equilibrium law of motion from initial capital stocks on both sides of K_A^{cs} , recovering the full conditional saddle in the phase diagram.

Figure 1: Conditional saddle paths and the phase diagram



Notes: Conditional K -nullclines ($\Delta K = 0$) partition the (K, C) plane. Direction arrows show frozen-regime convergence along each conditional saddle. Panel (a) zooms into the neighborhood of the conditional steady states shown in panel (b). The histogram plots the time-series distribution of aggregate capital.

decline along them; positive shocks reverse the pattern. Compared to time-domain representations, the phase diagram illustrates the full equilibrium dynamics more concisely in a single figure and enables immediate consideration of counterfactual scenarios. For the GIRF illustration, I replace the two-state Markov chain with a finely discretized AR(1) process for log productivity: $\ln A' = \rho \ln A + \epsilon$, $\epsilon \sim N(0, \sigma^2)$. In this case, there is a continuum of frozen- A conditional saddles indexed by A , and the GIRF path moves both along a given conditional saddle (endogenous propagation) and across the foliation as A evolves.

Figures 3 and 4 illustrate post-shock dynamics following a positive TFP shock. Upon impact, consumption jumps upward across conditional saddles. In subsequent periods, forces along and across saddles jointly generate a bow-shaped trajectory in the phase diagram: capital exhibits a hump-shaped response as the shock mean-reverts, while consumption initially rises then falls as TFP crosses the threshold where the direction of along-saddle adjustment reverses.

Comparison with the perfect-foresight saddles In the *perfect-foresight* (PF) economy, agents believe the current regime persists forever; the PF steady states solve $1 = \beta(1 - \delta + \alpha A(K_A^{pf})^{\alpha-1})$. Along the *conditional saddle* (CS), the regime is held fixed ex post, but agents correctly anticipate regime switches under Π , so the conditional steady states K_A^{cs} are pinned by the probability-weighted Euler equations.⁷ PF and CS economies share the same K -nullcline (Proposition 3), but differ in their consumption dynamics.

Proposition 3 (K -nullcline invariance over beliefs).

⁷Specifically, $\beta\pi_{AA}(1 + \alpha A(K_A^{cs})^{\alpha-1} - \delta) + \beta\pi_{AA'}(c(K_A^{cs}, A')/c_A^{cs})^{-1}(1 + \alpha A'(K_A^{cs})^{\alpha-1} - \delta) = 1$ for each $A \in \{B, G\}$, where $A' \neq A$.

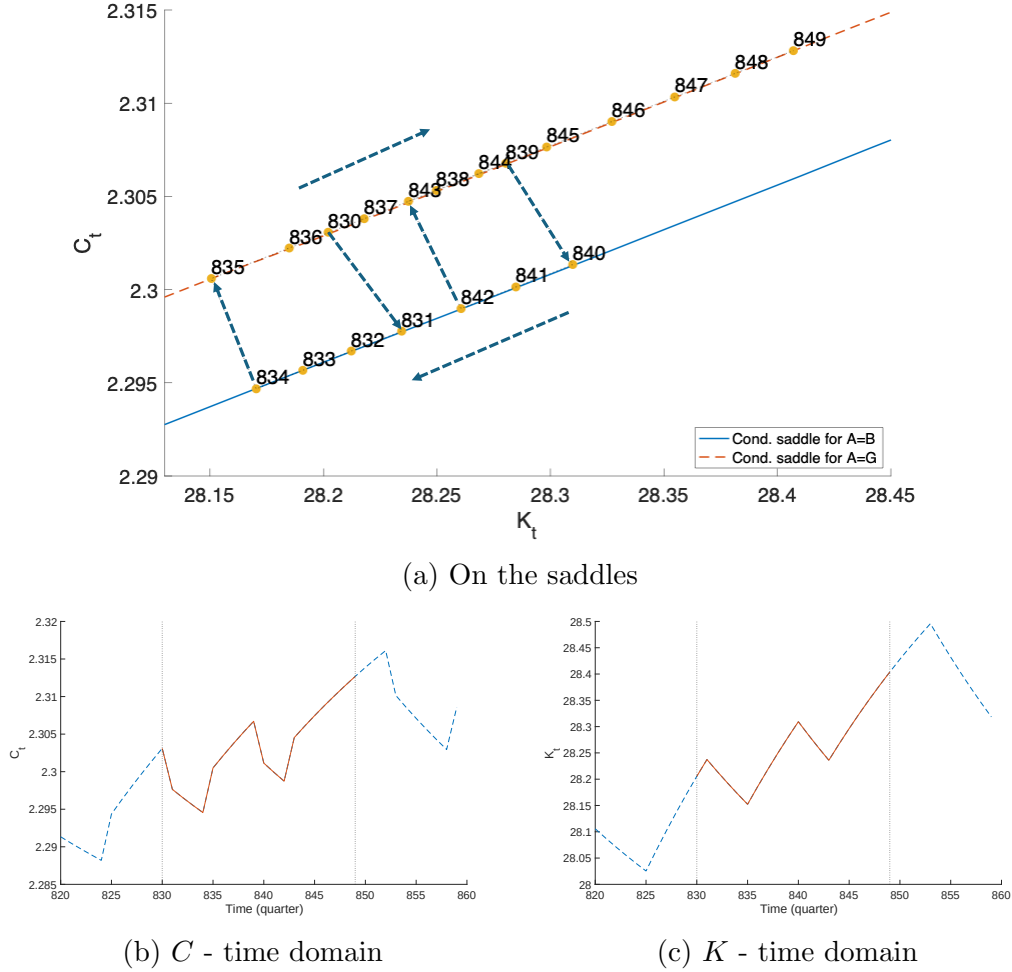


Figure 2: Equilibrium dynamics: saddle vs. time domain

Notes: Stochastic equilibrium dynamics on the conditional saddle paths (panel (a)) and in the time domain (panels (b) and (c)). Sample: 20 quarters (830–849).

Conditional K -nullclines are identical between the RBC model with aggregate uncertainty and the perfect-foresight counterpart.

Proof.

From the stationary condition $\delta K = I$ and the aggregate resource constraint $Y = C + I$, equation (16) is immediate. Neither condition involves expectations or belief structures, so the nullcline is identical whether or not there is aggregate uncertainty. ■

This invariance implies that PF–CS differences are driven entirely by consumption dynamics, not by the feasibility locus.⁸ Proposition 4 formalizes this via steady-state orderings.

⁸There is no consumption nullcline under aggregate uncertainty; the Euler equations at the conditional steady states play that role.

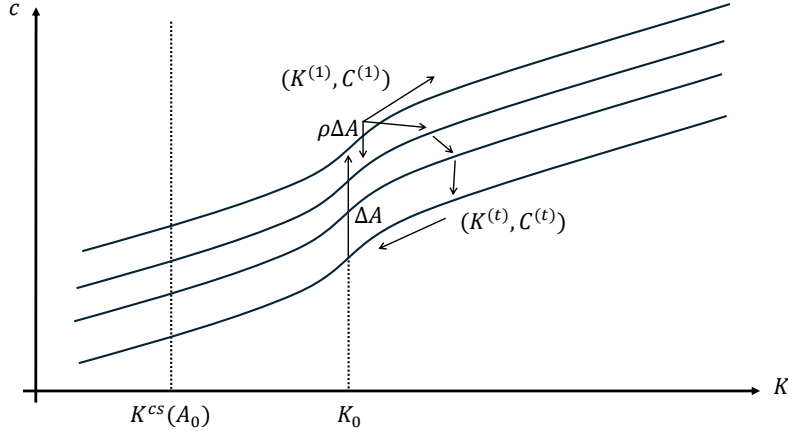


Figure 3: Generalized impulse response function (GIRF) at (K_0, A_0)

Notes: Generalized impulse responses of consumption and capital to a positive TFP shock at initial state (K_0, A_0) , where A_0 is the pre-shock productivity level. Upon impact, the economy jumps across conditional saddles to the $A_0 + \Delta$ foliation and subsequently propagates along and across saddles as TFP mean-reverts.

The same invariance extends to the heterogeneous-household environment (Section 4), where it underlies both K -sufficiency and the perfect-foresight diagnostic (Section 4.1).

Proposition 4 (Aggregate uncertainty and the conditional steady states).

Let $R(A, K) := 1 - \delta + \alpha AK^{\alpha-1}$ denote the gross return and $c(K, A')$ the stochastic-equilibrium consumption function. Suppose that (i) at each regime's PF steady state, the consumption response to a regime switch dominates the return response,

$$\frac{c(K_B^{pf}, G)}{c_B^{pf}} > \frac{R(G, K_B^{pf})}{R(B, K_B^{pf})}, \quad \frac{c(K_G^{pf}, B)}{c_G^{pf}} < \frac{R(B, K_G^{pf})}{R(G, K_G^{pf})}; \quad (18)$$

and (ii) each conditional Euler residual F_A —the residual of the conditional steady-state Euler equation viewed as a function of K (Appendix C)—is strictly positive below K_A^{cs} and strictly negative above it. Then

$$K_B^{cs} < K_B^{pf} < K_G^{pf} < K_G^{cs}, \quad c_B^{cs} < c_B^{pf} < c_G^{pf} < c_G^{cs}. \quad (19)$$

Proof.

See Appendix C. ■

Both hypotheses are explicit and directly checkable on the computed equilibrium. The return ratio in (18) deviates from one only at second order in the regime gap, $R(G, K)/R(B, K) =$

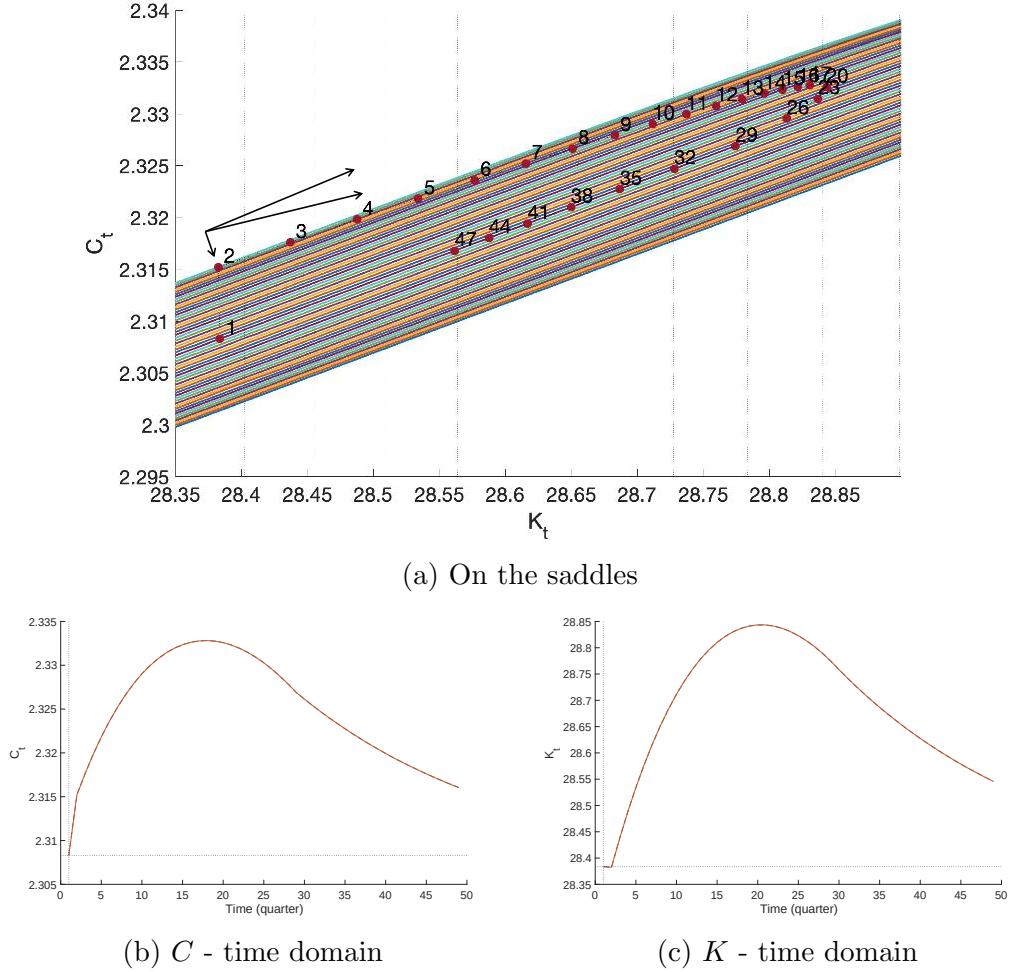


Figure 4: Impulse responses to a positive TFP shock: saddle vs. time domain

Notes: Impulse responses on conditional saddle paths (panel (a)) and in the time domain (panels (b) and (c)). Sample: 50 quarters.

$1 + (G - B) \alpha K^{\alpha-1} / R(B, K)$, while equilibrium consumption responds at first order to a persistent regime switch, so condition (i) fails only in degenerate parameterizations. Indeed, the conditions are tight: under full depreciation ($\delta = 1$), decision rules collapse to the belief-independent Brock–Mirman form, condition (i) holds with equality, and the conditional and PF steady states exactly coincide, so the strict ordering cannot hold unconditionally (Appendix C). Away from this corner, the orderings are confirmed in the computed equilibrium of Figure 5.

PF steady states are nested within the CS steady states: hedging against regime switches pushes the conditional steady states apart (Deaton, 1991; Carroll, 1997). Conversely, the conditional saddle *paths* are nested within the PF paths: at any given K , hedging compresses the consumption gap across regimes. The two nestings reflect the same mechanism: compressed

consumption responses cumulate into amplified long-run capital divergence (Figure 5).

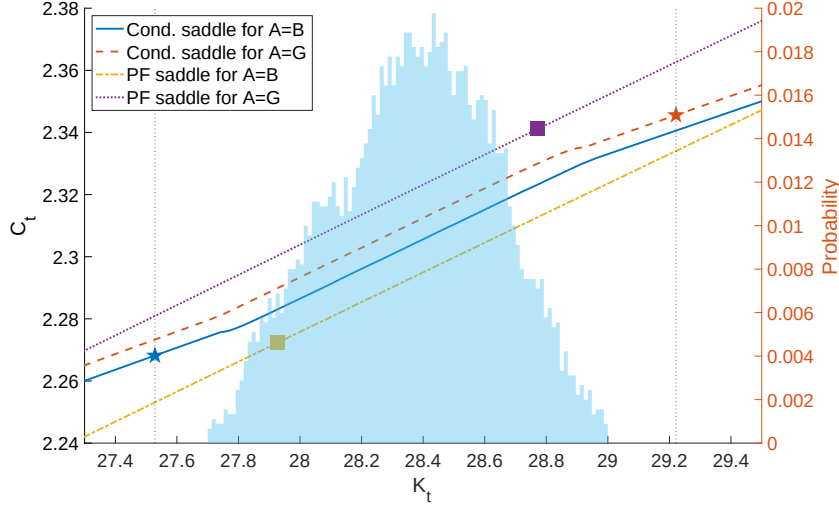


Figure 5: Conditional saddle path comparison: with and without uncertainty

Notes: Conditional saddle paths for $A = B$ (solid) and $A = G$ (dashed) and perfect-foresight saddle paths for $A = B$ (dash-dotted) and $A = G$ (dotted) under standard quarterly calibration.

Conditional boundary condition I work directly with the recursive formulation, which implicitly imposes a stochastic analogue of the deterministic transversality condition.⁹ Given conditional saddle paths, the natural counterpart is *conditional transversality*:

$$\lim_{t \rightarrow \infty} \beta^t u'(c_t) K_t = 0 \quad \text{under a frozen aggregate regime } A \in \{B, G\} \quad (20)$$

(along the conditional saddle for each $A \in \{B, G\}$).

Condition (20) rules out explosive continuations along each conditional saddle. Under standard assumptions, it is equivalent to the usual stochastic transversality conditions (Kamihigashi, 2003, 2005) and selects the same bounded RCE. The conditional formulation makes explicit that the boundary condition operates regime-by-regime on $\mathcal{M}(\Phi_0, A)$.

Economies without conditional saddle paths A conditional saddle requires a predetermined endogenous state. Without one, the economy jumps each period to the allocation implied by the current exogenous state, so the conditional “saddle” under frozen A collapses

⁹This condition is imposed at the aggregate level and should be distinguished from individual no-Ponzi constraints.

to a singleton. I call such an economy *endogenously stateless*.¹⁰ Models with predetermined states—capital, habits, interest-rate smoothing, or distributional states in HANK—are endogenously stateful, and conditional saddle paths exist under mild regularity conditions.

3 State dependence

In this section, I analyze nonlinear shock responsiveness through the lens of conditional saddle paths. In any stochastic dynamic model that admits conditional saddle paths, the response of an aggregate variable to an exogenous shock is represented by a shift across different conditional policy graphs. The natural benchmark is *state independence*: the impact gap between policy graphs is constant in the endogenous coordinate. Departures from this benchmark show up locally as slope differences between the graphs, which serve as the first-order diagnostic of state-dependent responses.

A large empirical literature documents that macroeconomic responses to shocks depend on the phase of the business cycle—recessions amplify negative shocks, expansions dampen them (Neftci, 1984; Auerbach and Gorodnichenko, 2012; Tenreyro and Thwaites, 2016). The following result formalizes the benchmark and its local failure.

Lemma 1 (Vertical-separation benchmark).

Let I be an interval over which the regime-specific conditional policy graphs $\mathcal{P}_{A_j}^g = \{(e, g_j(e)) : e \in I\}$, $j \in \{0, 1\}$, are single-valued, and define the impact gap $\Delta_g(e; A_1, A_0) := g_1(e) - g_0(e)$. Then Δ_g is constant on I if and only if $\mathcal{P}_{A_1}^g$ is a vertical translation of $\mathcal{P}_{A_0}^g$; if g_0, g_1 are differentiable on I , both are equivalent to the slope gap $g'_1 - g'_0$ vanishing on I .

Proof. The first equivalence is definitional; the second is the mean value theorem applied to Δ_g . ■

Lemma 1 is the elementary benchmark; the substantive statement—that the slope gap reflects model primitives—is Proposition 5 below. Sharp state independence is a knife-edge. Differential tilt across conditional policy graphs—whether from real (Winberry, 2021; Lee, 2025), financial (Melcangi, 2024), or labor market frictions (Petrosky-Nadeau et al., 2018; Pizzinelli et al., 2020)—generates state-dependent shock responses. Lemma 1 provides the qualitative characterization; the following proposition translates it into a structural diagnostic computable from model primitives. In the log-linearized frozen-regime RBC dynamics near

¹⁰The textbook three-equation New Keynesian model (Gali, 2008) under determinacy is endogenously stateless: $(x_t, \pi_t, i_t) = \Psi s_t$ for exogenous state $s_t := (r_t^n, u_t, \nu_t)$, so freezing s_t produces no nontrivial dynamics.

K_A^{cs} , let $R_A^{cs} := 1 + f'(K_A^{cs}; A) - \delta$ denote the gross return at the conditional steady state, $\lambda_A^s \in (0, 1)$ the stable eigenvalue, and $\psi_A := dC/dK|_{\text{saddle}, A}$ the level saddle slope at K_A^{cs} .

Proposition 5 (Slope ratio as a structural diagnostic).

The level saddle slope admits the closed form

$$\psi_A = R_A^{cs} - \lambda_A^s, \quad (21)$$

and hence the cross-regime slope ratio is

$$\mathcal{R} := \frac{\psi_B}{\psi_G} = \frac{R_B^{cs} - \lambda_B^s}{R_G^{cs} - \lambda_G^s}. \quad (22)$$

In log units the slope is $(K_A^{cs}/C_A^{cs})\psi_A$; both R_A^{cs} and λ_A^s are computable from primitives at the regime-specific conditional steady states.

Local smoothness used in slope analysis Definition 1 introduces the conditional saddle path $\mathcal{M}(\Phi_0, A)$ as an orbit closure—generically a countable orbit together with its limit point—rather than a smooth curve. The slope, curvature, and “tangent” language used in Proposition 5 (and subsequently in Proposition 6, the tangent-coherence discussion of Section 4, and Proposition 10) refers to the local C^1 parametrization of $\mathcal{M}(\Phi_0, A)$ near K_A^{cs} , which exists under the regularity conditions of Proposition 10 (Appendix F). Outside this neighborhood, the orbit is treated as the discrete object on which Theorem 1 guarantees injectivity.

Proof. Log-linearize the frozen-regime budget constraint $K_{t+1} = f(K_t; A) + (1 - \delta)K_t - C_t$ around (K_A^{cs}, C_A^{cs}) :

$$\hat{k}_{t+1} = R_A^{cs} \hat{k}_t - \frac{C_A^{cs}}{K_A^{cs}} \hat{c}_t. \quad (23)$$

On the stable manifold, $\hat{c}_t = \psi_A^{\log} \hat{k}_t$ for some log slope $\psi_A^{\log}$. Substituting into (23) yields $\hat{k}_{t+1} = [R_A^{cs} - \frac{C_A^{cs}}{K_A^{cs}} \psi_A^{\log}] \hat{k}_t$, which by definition equals $\lambda_A^s \hat{k}_t$. Solving, $\psi_A^{\log} = (K_A^{cs}/C_A^{cs})(R_A^{cs} - \lambda_A^s)$. Converting from log to level units via $dC/dK|_{K^{cs}} = (C_A^{cs}/K_A^{cs})\psi_A^{\log}$ gives (21). Equation (22) follows by taking the ratio.

Under perfect-foresight expectations, the steady-state Euler condition gives $R_A^{cs} = 1/\beta$ and the formula reduces to $\psi_A = 1/\beta - \lambda_A^s$. Under stochastic expectations over regime transitions, R_A^{cs} is pinned down by $1 = \beta \mathbb{E}_A[(C_A^{cs}/C(\Phi_A^{cs}, A'))^\sigma (1 + f'(K_A^{cs}; A') - \delta)]$ —the return is evaluated at tomorrow’s capital K_A^{cs} under the realized regime A' —and generally

differs from $1/\beta$; the formula remains valid with R_A^{cs} evaluated at the (stochastic) conditional steady state. ■

Proposition 5 gives the slope ratio $\mathcal{R}$ a structural interpretation: each regime’s saddle slope is the gap between the gross return on capital at that regime’s conditional steady state and the corresponding stable eigenvalue. Local departures from $\mathcal{R} = 1$ are the geometric signature of state-dependent impact responses, and can be computed *without* a global solution: one linearizes the frozen-regime dynamics around each conditional steady state, reads off (R_A^{cs}, λ_A^s) , and applies (22).

This diagnostic connects the geometric framework to the empirical state-dependence literature: $\mathcal{R}$ is a model-implied moment that could in principle be identified from state-dependent local projections (Auerbach and Gorodnichenko, 2012; Tenreyro and Thwaites, 2016).

An extension: asymmetric adjustment cost I modify the budget constraint by adding a quadratic wealth adjustment cost $\mathcal{C}(a', a) = \frac{\tilde{\mu}}{2} \left(\frac{a' - a}{a} \right)^2 a$, with $\tilde{\mu} = \mu_+$ for positive investment and $\tilde{\mu} = \mu_-$ for negative investment ($\mu_+ > \mu_-$).¹¹ Figure 6 plots the resulting conditional saddle paths. The $A = B$ saddle is substantially steeper than the $A = G$ saddle: $\mathcal{R} = \psi_B/\psi_G \approx 1.83$ ($\psi_B \approx 0.26$, $\psi_G \approx 0.14$; so $|\mathcal{R} - 1| \approx 0.83$ vs. the $\mathcal{R} = 1$ benchmark under symmetric costs). The ratio is a local quantity, evaluated at each regime’s conditional steady state, where the B saddle is at its steepest; it therefore exceeds the average tilt difference visible over the full plotted range, along which the B saddle flattens away from its conditional steady state. A one-standard-deviation TFP shock generates a larger consumption response when capital is low (3.42%) than when it is high (2.42%).

This geometric representation formalizes the intuition that “climbing up is difficult, falling down is easy” (Petrosky-Nadeau et al., 2018). By Lemma 1, state-independent impacts correspond to constant vertical separation; local departures from that benchmark appear as differential tilt. Such deviations—whether from financial (Melcangi, 2024), labor market (Petrosky-Nadeau et al., 2018; Pizzinelli et al., 2020), or capital-adjustment frictions—provide a structural interpretation of evidence that macroeconomic policy transmits asymmetrically over the cycle (Auerbach and Gorodnichenko, 2012; Tenreyro and Thwaites, 2016; Jo and Zubairy, 2025). The slope ratio $\mathcal{R}$ of Proposition 5 is a structural diagnostic of state dependence, computable from the linearized frozen-regime dynamics without a global solution.

¹¹The adjustment-cost parameters are not calibrated. For illustrative purposes, I set $\mu_+/\mu_- = 4$ ($\mu_- = 1.22$, $\mu_+ = 4.88$). I use an annual calibration ($\beta = 0.96$, $\delta = 0.10$, $\alpha = 0.33$) and a two-state TFP process $A \in \{0.98, 1.02\}$ with persistence 0.90.

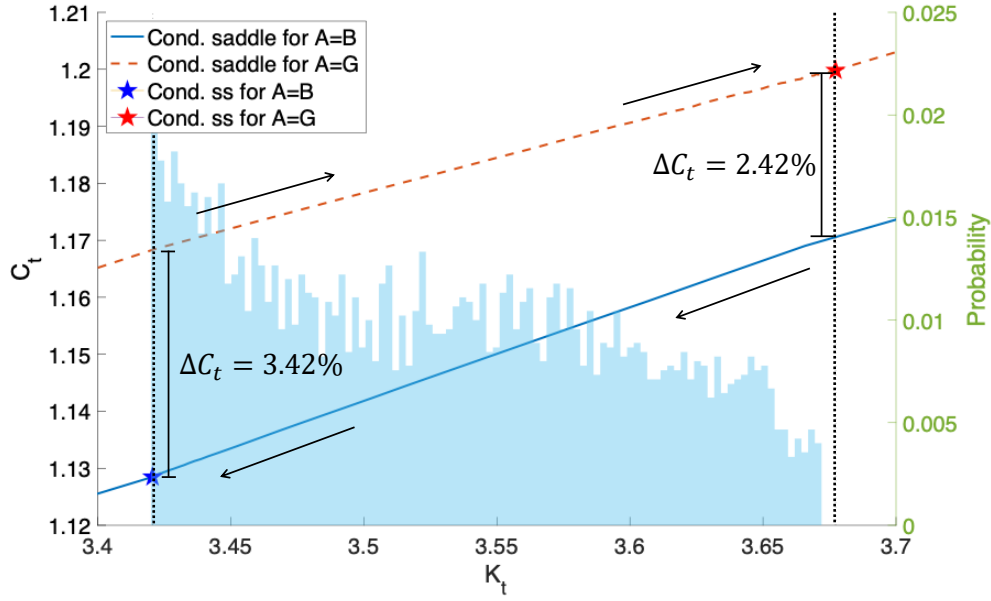


Figure 6: Differently tilted conditional saddle paths — endogenous state dependence

Notes: Conditional saddle paths for $A = B$ (solid) and $A = G$ (dashed) under asymmetric wealth adjustment cost ($\mu_+/\mu_- = 4$). The $A = B$ saddle is steeper, generating state-dependent shock responses.

Shape of the response function Even in the standard RBC, where $\mathcal{R} \approx 1$, the *shape* of the response trajectory is highly state dependent. A positive productivity shock displaces the economy across saddles, and the subsequent trajectory depends on K_0 's position relative to K_G^{cs} . Far from K_G^{cs} , the marginal product of capital is high, tilting the intertemporal tradeoff toward accumulation and producing a pronounced, late consumption hump (Barro and Sala-i Martin, 1992). Near K_G^{cs} , there is little room to accumulate; the across-saddle reversion dominates quickly and the peak arrives earlier. This channel is orthogonal to the slope asymmetry of Proposition 5. Slope differences govern *impact* magnitudes (cross-saddle dimension); position-dependent convergence speed governs response *shapes* (along-saddle dimension).

Crucially, this state-dependent shape does *not* arise from nonlinearity of the saddle path itself. Even in the log-linearized economy—where the conditional saddle is a straight line—peak timing varies with K_0 . The mechanism is the *speed of along-saddle convergence*. Far from K_G^{cs} , the capital gap is large relative to the shock-induced across-saddle displacement, so the along-saddle force (capital accumulation) dominates the across-saddle force (TFP reversion) for many periods before the balance tips, producing a later and more pronounced consumption hump. The shape asymmetry is thus a property of the *race between two convergence rates*—the endogenous along-saddle rate λ and the exogenous across-saddle rate ρ —and how initial distance from steady state weights the former relative to the latter.

Nonlinearity (concavity of $f(K)$) amplifies the effect quantitatively but is not its source.

The mechanism can be made precise via the log-linearized dynamics. Near the non-conditional steady state, the Blanchard–Kahn decomposition (Blanchard and Kahn, 1980) on the saddle branch yields

$$\hat{c}_t = \underbrace{\mathcal{P}(\hat{k}_0)}_{\text{along saddle}} \lambda^t + \underbrace{\mathcal{Q}}_{\text{across saddle}} \rho^t, \quad (24)$$

where $\hat{c}_t$ and $\hat{k}_0$ are log-deviations from steady state, $\lambda \in (0, 1)$ is the stable eigenvalue of the capital dynamics, $\rho \in (0, 1)$ is the TFP persistence, and $\varepsilon > 0$ is the initial shock size.¹² The coefficient $\mathcal{P} = \psi_k[\hat{k}_0 + \theta\varepsilon/(\lambda - \rho)]$ depends on initial capital through the saddle slope $\psi_k > 0$ and the capital-TFP elasticity $\theta > 0$. The coefficient $\mathcal{Q} = [\psi_a - \psi_k\theta/(\lambda - \rho)]\varepsilon$ does not. The hump shape arises from the tension between two decaying exponentials. The λ^t term captures along-saddle propagation—the endogenous capital accumulation effect—while the ρ^t term captures the direct TFP reversion. The peak occurs where these forces balance. A lower initial capital stock reduces $\mathcal{P}(\hat{k}_0)$, weakening the slowly decaying along-saddle term relative to the TFP-reversion term, so the balance tips later in time—exactly the comparative static $\partial t^*/\partial \hat{k}_0 < 0$ established below.

Proposition 6 (State-dependent peak timing).

Suppose $\lambda > \rho$, $\mathcal{P}(\hat{k}_0) > 0$, and the consumption response (24) admits a peak at $t^* > 0$. Then

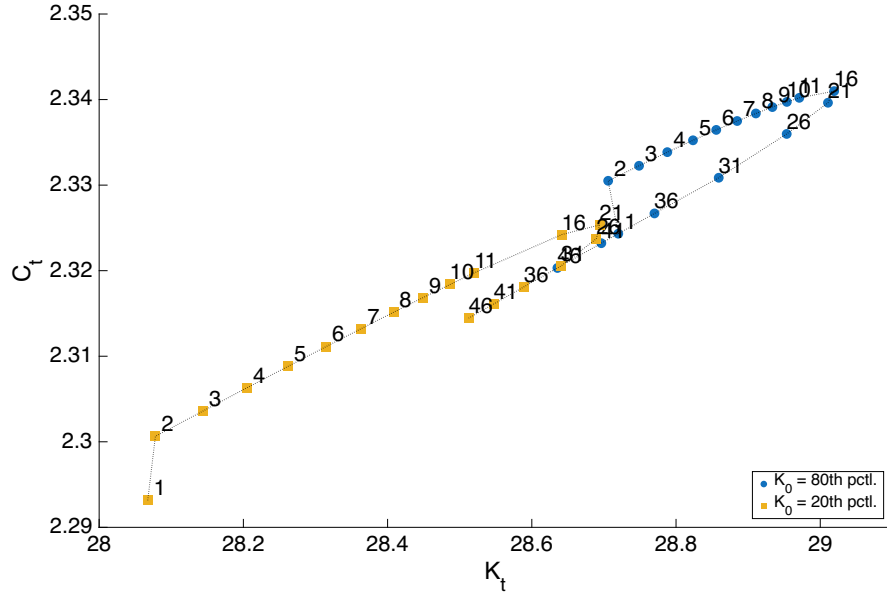
$$\frac{\partial t^*}{\partial \hat{k}_0} = \frac{-\psi_k}{\mathcal{P}(\hat{k}_0) \cdot \ln(\lambda/\rho)} < 0. \quad (25)$$

Proof.

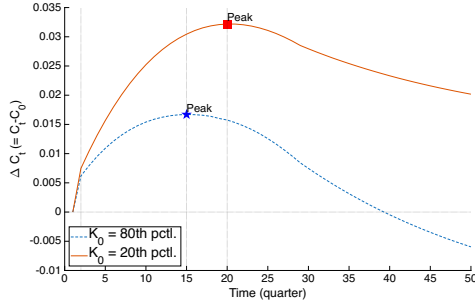
See Appendix D. ■

The consumption peak arrives strictly later when initial capital is farther below the steady state. Both hypotheses are mild: $\lambda > \rho$ holds under standard quarterly RBC calibrations ($\lambda \approx 0.96$, $\rho \leq 0.95$), and $\mathcal{P}(\hat{k}_0) > 0$ requires only $\hat{k}_0 > -\theta\varepsilon/(\lambda - \rho)$ —initial capital not farther below steady state than the long-run capital response to the shock—which the ergodic distribution satisfies with substantial margin. Proposition 6 is local by construction; Figure 7 confirms that the mechanism operates globally across the ergodic capital distribution.

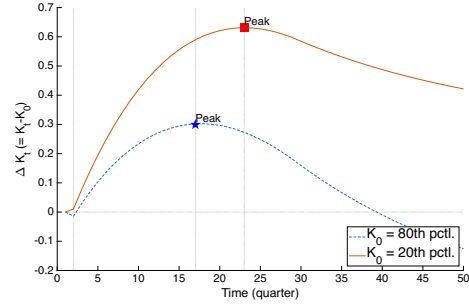
¹²This λ comes from the linearization at the *non-conditional* steady state and differs from the regime-specific λ_A^s of Proposition 5, which is linearized at the *conditional* steady state K_A^{cs} . To first order in the TFP perturbation, $\lambda_B^s = \lambda_G^s = \lambda$ in the frictionless RBC, consistent with $\mathcal{R} \approx 1$; friction-induced departures from this equality are what generate $\mathcal{R} \neq 1$. Proposition 6 operates in the $\mathcal{R} \approx 1$ regime, so its state dependence is driven by initial location $\hat{k}_0$, not by regime-varying slopes — an orthogonal channel.



(a) On the saddles



(b) C — time domain



(c) K — time domain

Figure 7: Shape of the impulse responses: low K_0 vs. high K_0

Notes: Impulse responses to a positive productivity shock from K_0 at the 20th percentile and 80th percentile of the ergodic distribution. Panel (a): post-shock trajectories in (K, C) space on the conditional saddle paths; numbers indicate quarters since impact. Panels (b) and (c): time-domain responses of consumption and capital; vertical markers indicate the peak quarter. Sample: 50 quarters.

Figure 7 confirms the logic of Proposition 6 quantitatively from the 20th and 80th percentiles of the ergodic capital distribution. From low K_0 , both C and K rise longer before reversion dominates, producing a later and larger consumption peak. From high K_0 , the peak arrives earlier—the same saddle geometry, the same shock, the same preferences; only the inherited capital stock differs.

4 Heterogeneous-household economy and dimension reduction

In a heterogeneous-household economy, the natural state variable is the entire cross-sectional distribution of wealth—an infinite-dimensional object. Classical aggregation ([Gorman, 1961](#)) reduces dimensionality by restricting preferences, but these restrictions fail under uninsurable idiosyncratic risk and borrowing constraints. Yet practitioners routinely approximate the state by a single number, aggregate capital. The conditional saddle framework explains why this works without preference restrictions: if the distribution evolves monotonically along a one-dimensional equilibrium path, then aggregate capital is an exact coordinate for the distribution along that path. This section formalizes the argument.

I consider a continuum of ex-ante homogeneous households of unit measure. The recursive formulation of the households' problem is as follows:

$$v(a, z; \Phi, A) = \max_{c, a'} \log(c) + \beta \mathbb{E}v(a', z'; \Phi', A') \quad (26)$$

$$c + a' = a(1 + r(\Phi, A)) + w(\Phi, A)z \quad (27)$$

$$a' \geq 0 \quad (28)$$

$$\Phi' = \Gamma_{\text{endo}}(\Phi, A), \quad A' \sim \Gamma_{\text{exo}}(A'|A). \quad (29)$$

The problem is the same as in the representative-household economy except for (i) uninsurable idiosyncratic labor productivity, which follows a Markov process $z \sim \Gamma_z(z'|z)$; (ii) inclusion of the distribution of individual states Φ in the aggregate endogenous state; and (iii) the corresponding change in the law of motion for the aggregate endogenous state. The RCE is defined as in [Krusell and Smith \(1998\)](#).

The model features two exogenous stochastic processes: idiosyncratic productivity and aggregate TFP. Accordingly, there are two layers of conditional saddle paths: individual and aggregate. The individual saddle path has its own cross-sectional implications, which are analyzed separately in Appendix B. Here I focus on the aggregate conditional saddle path under stochastic TFP, following [Krusell and Smith \(1998\)](#).¹³ Consistent with the notation in Section 2.2, I denote $K = K(\Phi)$ and $C = C(\Phi, A)$ as aggregate capital and consumption.

The conditional K -nullcline of Definition 5 extends directly to the heterogeneous-household setting: although the endogenous state is now the distribution Φ , net investment depends on Φ only through the aggregates (K, C) , so the locus remains an object in the aggregate phase

¹³In the original model of [Krusell and Smith \(1998\)](#), exogenous individual labor supply co-moves with aggregate TFP. All results are unaffected by this feature, but for expositional brevity I assume labor supply is exogenously fixed.

plane.

Definition 6 (Conditional K -nullcline in the heterogeneous-household setting).

Fix $A \in \{B, G\}$. The conditional K -nullcline is the zero-net-investment locus in the aggregate (K, C) plane,

$$\mathcal{N}(A) := \{(K, C) \in \mathbb{R}_{++} \times \mathbb{R}_{++} : \Delta K(K, C; A) = 0\},$$

where $\Delta K(K, C; A)$ is the one-period change in aggregate capital implied by the aggregate resource constraint at (K, C) under regime A . As in the representative-agent case, $\mathcal{N}(A)$ is distinct from the realized crossings $\{(K_t, C_t) : K_{t+1} = K_t\}$ along any frozen-regime continuation $\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A)$, which under strict K -monotonicity contain no point at all before the limit.

Krusell and Smith (1998) posits an endogenous law of motion that tracks aggregate capital K rather than the full distribution Φ : $\log K' = b_0(A) + b_1(A) \log K$, where b_0 and b_1 are state-dependent coefficients. This formulation embeds two distinct assumptions: (i) aggregate capital K suffices to summarize the endogenous aggregate state, and (ii) K follows a log-linear law of motion.

Since Krusell and Smith (1998), it has been an open question whether K -based forecasting works so well ($R^2 > 0.9999$) because of an accidentally accurate approximation or a deeper structural property.¹⁴ The conditional-saddle framework shows that (i) is exact within each frozen-regime spell: K uniquely indexes the distributional state on each conditional saddle path, so conditioning on K loses no information about within-spell dynamics. Proposition 9 below bounds the across-spell bridge to global accuracy—why a single forecasting rule $K' = f(K, A)$ works across all spells. The across-spell distributional discrepancy vanishes as regime persistence grows, providing a geometric foundation for the near-perfect computational accuracy. The same geometry indicates where approximate aggregation should strain: when nonconvexities flatten the nullcline gap or generate multiple K -stationary points—as in lumpy-investment economies where the cross-sectional distribution carries independent forecasting power (Khan and Thomas, 2008; Bachmann et al., 2013)—the monotonicity hypothesis weakens, and the framework predicts degraded scalar-state accuracy rather than assuming it away.

By contrast, (ii) remains a parametric approximation—the true law of motion for K is generally non-Markovian and need not be log-linear—but this is a matter of functional-form

¹⁴Den Haan (2010) shows that the R^2 of the forecasting regression can substantially overstate dynamic accuracy, because one-step-ahead fit masks the accumulation of errors over long simulated paths; his proposed accuracy test compares multi-period dynamic forecasts against the realized path. The across-spell handoff check reported below—comparing the globally solved K' against the (K, A) -based prediction at spell boundaries—is in the spirit of this stricter standard.

choice, not a limitation on the state space. The practical implication is that improving accuracy primarily requires richer functional forms for the law of motion, not additional distributional moments as state variables.

The following theorem provides the first step toward establishing the exact conditional sufficiency of K . The core insight is simple: if the economy converges along a one-dimensional equilibrium path and some aggregate variable moves strictly in one direction throughout, its current value tells you exactly where the economy is along that path. No two distinct distributional states can share the same capital level, because the economy passes through each level exactly once.

Theorem 1 (Monotone aggregate variable as a coordinate).

Fix (Φ_0, A_0) and $A \in \{B, G\}$. Let $\{\Phi_t\}_{t \geq 0}$ be the frozen-regime continuation $\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A)$ and let $\Phi^{cs} := \lim_{t \rightarrow \infty} \Phi_t$. Let $e : \mathcal{M}(\Phi_0, A) \rightarrow \mathbb{R}$ be an aggregate equilibrium variable. Suppose that the scalar sequence $\{e(\Phi_t)\}_{t \geq 0}$ is strictly monotone and converges to $e(\Phi^{cs})$. Then e uniquely indexes states on the conditional saddle:

$$e(\Phi) = e(\Phi') \implies \Phi = \Phi' \quad \forall \Phi, \Phi' \in \mathcal{M}(\Phi_0, A). \quad (30)$$

Proof. Define $\psi_A : \mathbb{N} \cup \{\infty\} \rightarrow \mathcal{M}(\Phi_0, A)$ by $\psi_A(t) := \Phi_t$ for $t \in \mathbb{N}$ and $\psi_A(\infty) := \Phi^{cs}$. By definition of $\mathcal{M}(\Phi_0, A)$ and convergence $\Phi_t \rightarrow \Phi^{cs}$,

$$\mathcal{M}(\Phi_0, A) = \{\Phi_t : t \geq 0\} \cup \{\Phi^{cs}\}, \quad (31)$$

so ψ_A is surjective. Strict monotonicity of $\{e(\Phi_t)\}$ implies that $e(\Phi_t) \neq e(\Phi_\tau)$ for $t \neq \tau$, hence $\Phi_t \neq \Phi_\tau$ for $t \neq \tau$; therefore ψ_A is injective on $\mathbb{N}$. Moreover, since $\{e(\Phi_t)\}$ is strictly monotone and converges to $e(\Phi^{cs})$, it also holds that $e(\Phi_t) \neq e(\Phi^{cs})$ for all finite t , so ψ_A is injective on $\mathbb{N} \cup \{\infty\}$. Thus ψ_A is a bijection between $\mathbb{N} \cup \{\infty\}$ and $\mathcal{M}(\Phi_0, A)$.

Now let $\Phi, \Phi' \in \mathcal{M}(\Phi_0, A)$. Pick $s, s' \in \mathbb{N} \cup \{\infty\}$ such that $\Phi = \psi_A(s)$ and $\Phi' = \psi_A(s')$. If $e(\Phi) = e(\Phi')$, then $e(\psi_A(s)) = e(\psi_A(s'))$. Since $e \circ \psi_A$ is injective, it follows that $s = s'$, hence $\Phi = \Phi'$. Therefore e is injective on $\mathcal{M}(\Phi_0, A)$, and an inverse map φ_A exists on $e(\mathcal{M}(\Phi_0, A))$. ■

Consequently, there exists a unique inverse map $\varphi_A : e(\mathcal{M}(\Phi_0, A)) \rightarrow \mathcal{M}(\Phi_0, A)$ such that $\varphi_A(e(\Phi)) = \Phi$ on $\mathcal{M}(\Phi_0, A)$, and any equilibrium object restricted to $\mathcal{M}(\Phi_0, A)$ can be written as a (single-valued) function of the scalar coordinate e .

The geometric intuition is that the conditional saddle is a one-dimensional curve of equilibrium states, and e acts as a “progress meter” along this curve: it moves strictly in one direction and never reverses. If two distinct states $\Phi \neq \Phi'$ shared the same value

$e(\Phi) = e(\Phi')$, both would be at the same “progress” position, yet their subsequent histories could not merge on a one-dimensional path—a contradiction. Hence e uniquely labels states on the conditional saddle.

The following proposition shows that the conditional K -nullclines coincide with those of the representative-agent model.

Proposition 7 (Conditional K -nullclines of Krusell and Smith (1998)).

The conditional K -nullclines of the heterogeneous-household model are identical to those of the representative-household model and invariant over the initial distribution Φ_0 :

$$C_A^{Knull}(K) = AK^\alpha - \delta K, \text{ for } A \in \{B, G\}. \quad (32)$$

Proof.

As in Proposition 3, the stationary condition $\delta K = I$ and the aggregate resource constraint $Y = C + I$ immediately imply the form of the conditional nullclines. ■

The nullcline is pinned by the aggregate resource constraint—an accounting identity indifferent to *who* saves. At any K , output AK^α and depreciation δK are determined by technology alone, so the nullcline $C = AK^\alpha - \delta K$ is distribution-free. This is what enables dimension reduction: the “obstacle” governing the sign of capital accumulation does not move as the distribution evolves along the conditional saddle.

The invariance is specific to the neoclassical production structure with exogenous labor supply. Under separable CRRA preferences with wealth effects on labor, labor supply introduces a distributional channel into the resource constraint. The nullcline then acquires distributional dependence, blocking the analytical route to K -sufficiency (though the perfect-foresight diagnostic of Section 4.1 remains applicable).

Proposition 8 (K monotonicity and injectivity).

Fix $A \in \{B, G\}$. Under Assumptions 1 and 2, together with unique K -stationarity along the frozen- A continuation (the no-plateau condition of Section 2.2), aggregate capital converges to K_A^{cs} strictly monotonically on $\mathcal{M}(\Phi_0, A)$. Consequently, K is injective on $\mathcal{M}(\Phi_0, A)$.

Proof.

By Proposition 7, the conditional K -nullcline takes the same form as in the RBC benchmark. The proof of strict monotonicity therefore follows that of Proposition 2 verbatim, with the RBC nullcline replaced by $C_A^{Knull}(K) = AK^\alpha - \delta K$ and with unique K -stationarity supplied locally by the stable-eigenvalue condition and maintained globally as a pathwise hypothesis, verifiable by direct inspection of the simulated frozen-regime continuation. The primitive MPC condition (Appendix G, Lemma G.1) constrains the no-side-switching property along

the continuation but does not on its own rule out non-steady-state flat steps. Since $\{K_t\}$ is strictly monotone and converges to K_A^{cs} , applying Theorem 1 with $e(\Phi) = K(\Phi)$ yields $K(\Phi) = K(\Phi') \implies \Phi = \Phi'$ for all $\Phi, \Phi' \in \mathcal{M}(\Phi_0, A)$. ■

Proposition 7 makes the K -nullcline *sign-determining*: the graph $C = C_A^{Knull}(K)$ partitions the (K, C) plane into regions with $\Delta K > 0$ and $\Delta K < 0$, via $\Delta K = C_A^{Knull}(K) - C$. Assumption 2 rules out discrete-time jumps across this partition by requiring that the segment connecting (K_t, C_t) and (K_{t+1}, C_{t+1}) does not intersect the nullcline. Together, these two ingredients imply that net investment cannot reverse sign along the conditional saddle. Capital dynamics are therefore strictly monotone, and each frozen-regime spell admits an exact one-dimensional representation indexed by K .

Proposition 8 says that, conditional on (Φ_0, A) , the mapping $K : \mathcal{M}(\Phi_0, A) \rightarrow \mathbb{R}$ is one-to-one, so an inverse map $\varphi_{\Phi_0, A}$ exists with $\Phi = \varphi_{\Phi_0, A}(K(\Phi))$ on $\mathcal{M}(\Phi_0, A)$.¹⁵ Any equilibrium object restricted to $\mathcal{M}(\Phi_0, A)$ therefore admits an exact single-valued representation as a function of (K, A) .

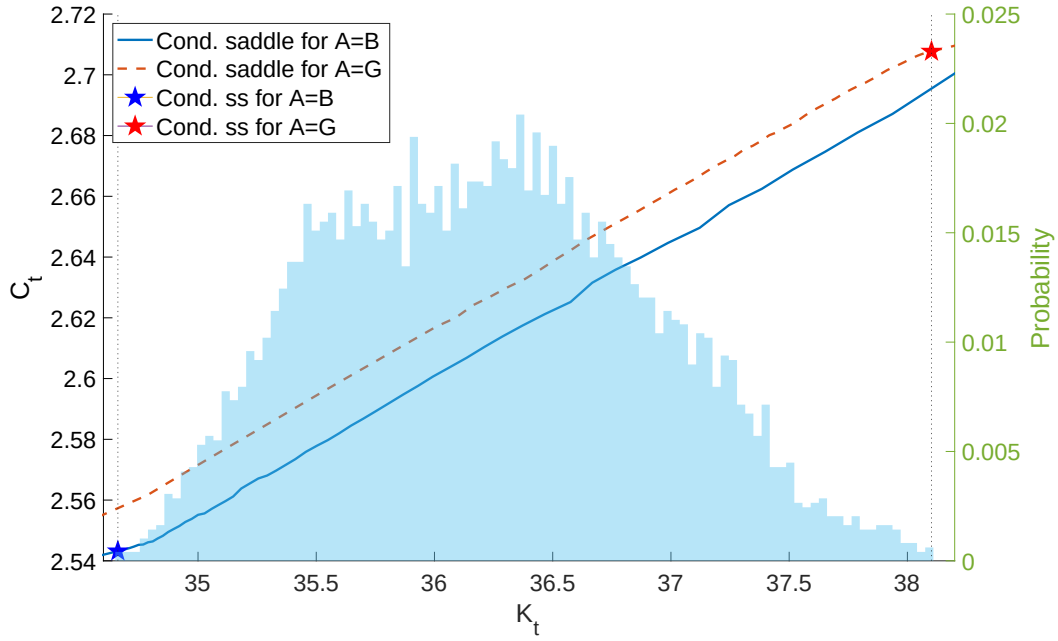


Figure 8: Conditional saddle paths in Krusell and Smith (1998)

Notes: The figure plots conditional saddle paths for $A = B$ (solid) and $A = G$ (dashed) implied by a canonical heterogeneous household business cycle model (Krusell and Smith, 1998). The histogram in the background plots the time-series distribution of the aggregate capital stock.

The tight ergodic alignment in Figure 8 could mask the within-spell structure that

¹⁵This is a path-specific notion of sufficiency: for a fixed (Φ_0, A) , K uniquely indexes states on the entire conditional saddle (i.e., globally *along the transition path*).

Proposition 8 pins down. Figure 9 resolves this by zooming into a narrow K -window: the cloud, line-like at the ergodic scale, is visibly “dusty” once resolved, yet within each individual spell the consecutive (K_t, C_t) points (linked by dashed lines) move monotonically in K . Each spell traces out an exact one-dimensional orbit; ergodic mixing across spells produces the band of points one observes at any given K .

To complement this theoretical result, Figure 8 plots the computed conditional saddle paths in the (K, C) phase diagram. The model is solved globally using the repeated transition method, and the dynamics under each frozen aggregate state are simulated for 2,500 periods. Although the true endogenous state of the model is the full distribution Φ rather than K alone, the figure shows that conditional saddle paths are strictly and monotonically ordered in K , providing clear computational support for the sufficiency result.

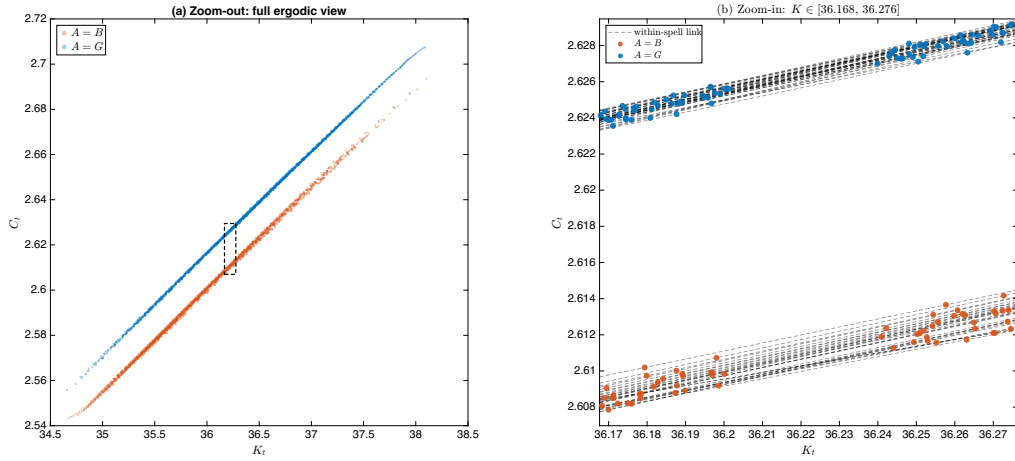


Figure 9: KS98 conditional saddle: zoom-out (left) vs. zoom-in (right)

Notes: Panel (a) shows the full ergodic view; the dashed rectangle marks the zoom window of panel (b). Panel (b) shows a narrow K -window ($\pm 0.15\%$ around the ergodic median); each ergodic observation is plotted as a colored marker (orange $A = B$, blue $A = G$), with consecutive within-spell observations linked by black dashed lines. Each dashed link is monotone in K , illustrating Proposition 8: within a spell, K uniquely indexes the distributional state. The visible cross-spell “dust” is precisely what Proposition 9 bounds.

A closer reading of Figure 9 reveals more structure than Proposition 8 alone delivers: the within-spell dashed links are not merely monotone in K but visibly *parallel*. This parallelism is the visual signature of *first-order* K -sufficiency—the local slope dC/dK along each conditional saddle is approximately invariant to the starting distributional condition Φ_0 . Strict K -sufficiency would require both zeroth-order coincidence (same C at common K) and first-order coincidence (same slope at common K). The bounded dust in the zoom panel is the (bounded) failure of the former, governed by Proposition 9; the parallelism is the latter holding approximately. Asymptotically this is Proposition 10 at work: near the

conditional steady state, all orbits align with a single direction, so their slopes converge to the common value $\psi_A = R_A^{cs} - \lambda_A^s$ regardless of starting point. A formal global version of this slope-coincidence claim would require additional geometric structure and is left to future work.

From spell-wise to global sufficiency Proposition 8 is stated regime-by-regime: K uniquely indexes the conditional saddle $\mathcal{M}(\Phi_0, A)$ under a single frozen regime. The equilibrium path is a concatenation of such one-dimensional segments, each exactly indexed by (K, A) , with the handoff between spells pinned down by the scalar pair (K_τ, A_τ) . Spell-wise sufficiency is distinct from unconditional global sufficiency, which would require the same K to pin the same Φ regardless of path history. The following proposition bounds the across-spell discrepancy.

The key intuition is as follows. When the economy is observed at a given K , different prior histories may have produced different underlying distributions Φ —a “cloud” of distributions at the same capital level. How tight is this cloud? The answer comes from a synchronization property of contracting random dynamics: when every frozen-regime map contracts, the state today is determined by the regime history alone, up to an error that decays geometrically in the length of history. Two states reached through histories that agree in recent periods must therefore be close, no matter how different their distant pasts. The dispersion in the same- K cloud is thus governed by how far back two histories must be traced before they diverge.

Write $\Gamma_A := \Gamma_{\text{endo}}(\cdot, A)$. Say that $\{\Gamma_A\}_{A \in \{B, G\}}$ is a *uniform frozen-regime contraction* on $\mathcal{U}$ at rate $\lambda_c \in (0, 1)$ if $\mathcal{U} \subseteq \mathcal{X}$ is closed in the complete metric space $(\mathcal{X}, d)$, forward-invariant under every Γ_A , of finite diameter $\text{diam}(\mathcal{U}) := \sup_{\Phi, \Phi' \in \mathcal{U}} d(\Phi, \Phi')$, and $d(\Gamma_A \Phi, \Gamma_A \Phi') \leq \lambda_c d(\Phi, \Phi')$ for all $\Phi, \Phi' \in \mathcal{U}$ and all A .

Theorem 2 (Synchronization under frozen-regime contraction).

Let $\{\Gamma_A\}$ be a uniform frozen-regime contraction on $\mathcal{U}$ at rate λ_c . Then:

(i) (Random fixed point) For every regime history $\omega = (\dots, A_{t-1}, A_t)$, the pullback limit

$$\Phi^*(\omega) := \lim_{T \rightarrow \infty} (\Gamma_{A_t} \circ \Gamma_{A_{t-1}} \circ \dots \circ \Gamma_{A_{t-T}})(\Phi)$$

exists, is independent of $\Phi \in \mathcal{U}$, and is the unique state in $\mathcal{U}$ consistent with ω ; any trajectory initialized in $\mathcal{U}$ at date $t - T$ satisfies $d(\Phi_t, \Phi^*(\omega)) \leq \lambda_c^T \text{diam}(\mathcal{U})$.

(ii) (Suffix synchronization) If two histories ω, ω' agree over their most recent T periods, then

$$d(\Phi^*(\omega), \Phi^*(\omega')) \leq \lambda_c^T \text{diam}(\mathcal{U}).$$

Proof.

See Appendix E. ■

Part (i) is the random-fixed-point construction of [Schenk-Hoppé and Schmalfuß \(2001\)](#) specialized to the frozen-regime maps; part (ii) is its consequence for any two states in the same- K cloud. The cloud bound follows by averaging part (ii) over the distribution of the common-suffix length. For the stationary symmetric two-state chain ($\pi_{BB} = \pi_{GG} = \rho_A$), let $q := \rho_A^2 + (1 - \rho_A)^2$ denote the probability that two independent histories agreeing at a date also agree one period earlier.¹⁶

Proposition 9 (Across-spell approximation bound).

Let $\{\Gamma_A\}$ be a uniform frozen-regime contraction on $\mathcal{U}$ at rate λ_c , and let $\Phi, \Phi' \in \mathcal{U}$ be the pullback limits of two regime histories that agree over their most recent $T \geq 1$ periods. Then

$$d(\Phi, \Phi') \leq \lambda_c^T \text{diam}(\mathcal{U}).$$

If, moreover, the two histories are drawn independently from the stationary symmetric chain, conditioned to share the current regime, then $T - 1$ is geometric with success probability $1 - q$ and

$$\mathbb{E}[d(\Phi, \Phi')] \leq \frac{(1 - q) \lambda_c}{1 - q \lambda_c} \text{diam}(\mathcal{U}).$$

Proof.

See Appendix E. ■

The diameter $\text{diam}(\mathcal{U})$ is finite under the natural borrowing limit and compact idiosyncratic support. Unlike a bound stated in terms of the same- K cloud itself, the right-hand side involves only the primitive set $\mathcal{U}$; the proposition is therefore a genuine bound on the cloud rather than a contraction statement about it. The bound vanishes as $\rho_A \rightarrow 1$ (long spells make recent suffixes long) or $\lambda_c \rightarrow 0$ (fast within-spell convergence).

Uniform contraction is a substantive assumption The hypothesis that $\Gamma_{\text{endo}}(\cdot, A)$ contracts uniformly on $\mathcal{U}$ is not innocuous: the map embeds general-equilibrium price feedback, and establishing contraction-type properties for such operators is precisely the difficulty confronted by global existence and uniqueness analyses ([Proehl, 2025](#)). Locally, Proposition 10 anchors the contraction rate to the stable eigenvalue $\lambda_A^s \in (0, 1)$; globally, the assumption

¹⁶Stepping backward one period, the two independent histories continue to agree if both persist (ρ_A^2) or both switch ($(1 - \rho_A)^2$); symmetry makes the time-reversed chain identical to the forward chain. For asymmetric chains, q is replaced by the corresponding backward agreement probability of the time-reversed chain.

is maintained, and the sample-based diagnostic below provides an empirical check—a lower-bound estimate, not a certification. In particular, the paper supplies no upper bound on λ_c : even the hypothesis $\lambda_c < 1$ is maintained rather than verified. No tension with the saddle structure arises, however: Γ_{endo} is the equilibrium law of motion after the forward-looking variable has been solved out, so the unstable direction of the (K, C) system is absent from the distributional dynamics.

Ergodic interpretation Theorem 2(i) is what licenses reading Proposition 9 on the ergodic set. Every state observed along a long simulated path is, up to a geometrically small error, the pullback limit of its own realized history—the state that history alone drives the economy toward, regardless of where it started; every pair in the same- K cloud is therefore a pair of pullback limits of regime histories and inherits the pathwise bound $d(\Phi, \Phi') \leq \lambda_c^T \text{diam}(\mathcal{U})$, with T the realized common-suffix length. This pathwise bound applies at every date—deep within a spell or just after a switch—because the synchronization argument runs over histories rather than over single spells. The expectation formula, by contrast, is computed under the idealized model of two independent stationary histories; within-path pairs are dependent and conditioning on a common K is a selection, so the closed form serves as a benchmark for the typical scale of the cloud rather than a bound on its realized diameter.

Theorem 2 leaves λ_c unspecified, only requiring that it bounds the per-period contraction on $\mathcal{U}$. The next proposition pins down what λ_c should be near the conditional steady state: it is just the stable eigenvalue of the linearized frozen-regime dynamics—the same object that governs the local saddle slope in Proposition 5.

Proposition 10 (Local contraction rate on the conditional saddle).

Near the conditional steady state, the frozen-regime dynamics on the conditional saddle contract distances at the geometric rate $|\lambda_A^s|$, where $\lambda_A^s \in (0, 1)$ is the stable eigenvalue of the linearized frozen-regime dynamics; in aggregate-capital units,

$$K_{t+1} - K_A^{cs} = \lambda_A^s (K_t - K_A^{cs}) + o(|K_t - K_A^{cs}|).$$

Proof.

See Appendix F. ■

The proposition is stated locally and along the conditional saddle; the full statement on the distributional state space, its regularity conditions (a C^1 Banach chart, C^1 Fréchet differentiability of Γ_{endo} , a simple stable eigenvalue with a spectral gap, a C^1 invariant curve containing the orbit tail, and metric compatibility), and the proof are in Appendix F. Directions orthogonal to the saddle can contract at different rates, so the global contraction

rate λ_c on $\mathcal{U}$ in Theorem 2 could in principle exceed $|\lambda_A^s|$. What the proposition does deliver is the right primitive anchor: the same eigenvalue that drives the slope-ratio diagnostic (Proposition 5) disciplines the local contraction entering Theorem 2 and Proposition 9—a single primitive underlies both. A certified uniform upper bound on λ_c over $\mathcal{U}$ requires separate machinery (Lipschitz constants or confidence bounds) and is left to future work; the sample-based diagnostic below provides a lower-bound estimate with $|\lambda_A^s|$ as a reality check.

A sample-based diagnostic for λ_c . The global uniform λ_c in Theorem 2 bounds contraction across a cloud of distributional states at the same K reached from different histories, which need not lie on a single conditional saddle. The across-spell bound uses λ_c as an *upper* bound on the per-step contraction rate, so a certified verification of Proposition 9 requires a certified upper bound on λ_c . Obtaining such an upper bound requires either a Lipschitz constant for Γ_{endo} on $\mathcal{U}$ (model-specific), or a statistical confidence bound layered onto sample estimates (standard but tedious in function spaces). A certified upper bound is beyond the scope of this paper and is left to future work. Instead, I propose a sample-based *diagnostic* that provides a lower bound on λ_c and a reality check against the local anchor $|\lambda_A^s|$.

The diagnostic proceeds in three steps. (i) Draw N pairs of distributional states $(\Phi^{(i)}, \Phi'^{(i)})$, $i = 1, \dots, N$, from the simulated equilibrium on $\mathcal{U}$, sampling the across-history cloud at common K values. (ii) Compute empirical contraction ratios

$$r_i := \frac{d(\Gamma_{\text{endo}}(\Phi^{(i)}, A), \Gamma_{\text{endo}}(\Phi'^{(i)}, A))}{d(\Phi^{(i)}, \Phi'^{(i)})}.$$

(iii) Report $\hat{\lambda}_c^N := \max_i r_i$. By construction, $\hat{\lambda}_c^N \leq \lambda_c$ —the sample max cannot exceed the true supremum—so $\hat{\lambda}_c^N$ is a *sample-based lower bound*, and substituting it for λ_c in the across-spell formula gives an *optimistic* approximation. Proposition 10 supplies the along-saddle anchor $|\lambda_A^s|$ as a reality check. When off-saddle dynamics contract at comparable rates, $|\lambda_A^s|$ and $\hat{\lambda}_c^N$ should agree closely, indirect evidence that the true λ_c is not much larger.

Under the [Krusell and Smith \(1998\)](#) calibration, quarterly regime persistence is $\rho_A = \pi_{AA} = 0.875$ (average 8-quarter spells) and the sample diagnostic yields $\hat{\lambda}_c^N \approx 0.964$, with the local anchor $|\lambda_A^s| \approx 0.963$ agreeing to three decimals. The backward agreement probability is $q = \rho_A^2 + (1 - \rho_A)^2 \approx 0.781$, and substituting $\hat{\lambda}_c^N$ into the benchmark formula of Proposition 9 gives $\mathbb{E}[\lambda_c^T] \approx 0.85$ as the typical contraction factor multiplying $\text{diam}(\mathcal{U})$ under the independent-histories benchmark—an optimistic estimate on both counts. The direct quantitative check—comparing the globally solved K' at each spell boundary against the prediction from tracking only (K, A) —gives an average discrepancy of 2.5×10^{-4} in log terms (1.3% of one standard deviation of $\ln K$), consistent with the sample-based estimate and

confirming that the across-spell error is small in this calibration.

From near rationality to complete rationality The [Krusell and Smith \(1998\)](#) approach is often labeled near-rational ([Akerlof and Yellen, 1985](#)), but this conflates two independent issues. Conditioning on K is exact within each spell—no two states within the same frozen-regime spell share the same capital level, so agents lose no within-spell information by tracking K alone. What *is* approximate is the log-linear functional form mapping today’s K to tomorrow’s. The bounded rationality, if any, resides in the parametric law of motion, not in the choice of state variable. Expanding the state space beyond K yields no within-spell improvement; enriching the functional form is the productive margin. The global accuracy ($R^2 > 0.9999$) reflects exact within-spell sufficiency and quantitatively small across-spell errors (Proposition 9).

Connection to block recursive equilibrium The conditional sufficiency of K is closely related to *block recursive equilibrium* (BRE), in which aggregate prices and quantities can be determined from a low-dimensional state without tracking the full distribution ([Menzio and Shi, 2010, 2011](#); [Schaal, 2017](#)). In the directed-search environments of [Menzio and Shi \(2011\)](#) and [Schaal \(2017\)](#), block recursivity holds exactly by construction: market segmentation severs the feedback from the distribution to individual decision problems. Here, by contrast, the structure is not built into the environment but emerges spell-wise from the saddle geometry. Within each frozen-regime spell, the conditional saddle framework delivers exactly this structure: K pins down all aggregate equilibrium objects, and individual policies are determined given (K, A) . Spell-wise K -sufficiency thus establishes that the equilibrium is block recursive *conditional on each regime*—aggregate prices, quantities, and the distributional state Φ itself are all recoverable from (K, A) on the conditional saddle path. Whether the equilibrium is globally block recursive depends on the across-spell discrepancy bounded in Proposition 9. Under the [Krusell and Smith \(1998\)](#) calibration, this discrepancy is quantitatively negligible.

Parametric form of the conditional saddle paths Despite the indexing function K , the specific form of the law of motion remains undetermined. The repeated transition method ([Lee, 2025](#)) exploits the recurrence of equilibrium allocations along the conditional saddle path; Theorem 1 and Proposition 8 provide the theoretical foundation for implementing the RTM using K as the conditional sufficient statistic in heterogeneous-agent models.

Verifying monotonicity in practice The sufficiency results above rely on strict monotonicity of K along frozen-regime continuations (Assumption 2). How should a researcher

verify this in a given model? The following two subsections develop complementary strategies. Section 4.1 provides a *perfect-foresight diagnostic*: one solves the deterministic ($\mu = 0$) transition dynamics and checks whether K converges monotonically; combined with a no-plateau condition and uniform spectral regularity along the homotopy, this transports monotonicity from $\mu = 0$ to $\mu = 1$ (rational expectations). Section 4.2 provides an independent route from static primitives—bounds on household MPCs and price feedback—that does not require solving any transition path.

4.1 Perfect-foresight diagnostic for monotonicity

This section shows that K -monotonicity under rational expectations can be verified from the deterministic perfect-foresight model, paired with two additional regularity conditions: unique K -stationarity (no non-steady-state flat steps) and uniform spectral regularity along the homotopy. The key insight is that the K -nullcline—the “obstacle” that the path must not cross—is invariant over beliefs (Proposition 7), so monotonicity can be transported continuously from perfect foresight to rational expectations once the regularity conditions are in place.

Belief family For $\mu \in [0, 1]$, define the belief transition matrix $\Pi(\mu) := (1 - \mu)I + \mu\Pi$, where I is the identity and Π is the true Markov matrix for $A \in \{B, G\}$. At $\mu = 0$, agents assign probability one to the current regime persisting (perfect foresight); at $\mu = 1$, agents hold rational expectations. For each μ , the μ -RCE is the recursive competitive equilibrium under $A' \sim \Pi(\mu)(\cdot|A)$ with the same technology and market structure. Denote the frozen-regime continuation under (μ, A) by $\{(K_t(\mu), C_t(\mu))\}_{t \geq 0}$, with conditional steady state $K_A^{cs}(\mu)$.

Under standard regularity, the frozen-regime continuation and conditional steady state are continuous in μ .¹⁷ The question is whether K -monotonicity, which may hold at $\mu = 0$ (perfect foresight), survives the passage to $\mu = 1$ (rational expectations). To formalize this, define the *monotonicity set*

$$\mathcal{S} := \{\mu \in [0, 1] : \{K_t(\mu)\}_{t \geq 0} \text{ is strictly monotone along the frozen-}A \text{ continuation}\}.$$

The next result anchors the verification on a single *deterministic* check—monotonic convergence of the perfect-foresight transition path, a textbook property of the Ramsey model—

¹⁷Since $\Pi(\mu)$ enters the Bellman equation linearly, the recursive equilibrium operator is continuous in μ . The result follows by the parametric contraction mapping theorem and continuous dependence of the market-clearing fixed point. See Appendix H.

paired with two regularity conditions along the homotopy: a no-plateau condition (unique K -stationarity) and uniform spectral regularity. Under these, monotonicity transports from perfect foresight to rational expectations. The intuition is that the K -nullcline is the boundary between the rising- K and falling- K regions of the phase diagram, and it does not move as beliefs vary from perfect foresight to rational expectations (Proposition 7). Suppose the perfect-foresight path stays on one side of this immovable boundary. Turning on rational expectations smoothly cannot push the path across it without violating the regularity conditions: a crossing would require the nullcline gap to pass through zero, which continuity rules out.

Call the belief homotopy *regular* if the family $\{(\Phi_t(\mu))_{t \geq 0}, \Phi_A^{cs}(\mu)\}_{\mu \in [0,1]}$ is jointly continuous in μ and, at every $\mu \in [0,1]$, the linearized frozen-regime operator at $\Phi_A^{cs}(\mu)$ has a simple stable eigenvalue $\lambda_A^s(\mu) \in (0,1)$ separated from the rest of the spectrum by a uniform spectral gap—the conditions of Proposition 10 and Appendix H, applied uniformly over the homotopy.

Proposition 11 (Perfect-foresight diagnostic).

Fix $A \in \{B, G\}$ and suppose: (i) the perfect-foresight ($\mu = 0$) transition path is strictly K -monotone; (ii) $\Delta K_t = 0$ only at $K = K_A^{cs}(\mu)$, for every $\mu \in [0,1]$; and (iii) the belief homotopy is regular. Then aggregate capital is strictly K -monotone along the rational-expectations ($\mu = 1$) frozen-regime continuation.

Proof.

Since $[0,1]$ is connected, it suffices to show $\mathcal{S}$ is nonempty, open, and closed.

Nonempty. $0 \in \mathcal{S}$ by assumption.

Open. Fix $\mu^* \in \mathcal{S}$ and define the nullcline gap $H_t(\mu) := C_t(\mu) - C_A^{Knull}(K_t(\mu))$, so that $\Delta K_t(\mu) = -H_t(\mu)$. At μ^* , strict monotonicity of K implies $H_t(\mu^*)$ has constant nonzero sign for all pre-limit t . By Proposition 7, the nullcline C_A^{Knull} is independent of μ . Combined with continuous dependence of $(K_t(\mu), C_t(\mu))$ on μ , for any finite T there exists $\varepsilon > 0$ such that $|\mu - \mu^*| < \varepsilon$ implies $H_t(\mu) H_t(\mu^*) > 0$ for all $t \leq T$.

For the tail: monotone convergence at μ^* implies the stable eigenvalue of the linearized frozen-regime dynamics at $K_A^{cs}(\mu^*)$ is real and lies in $(0,1)$ —otherwise local convergence would be oscillatory. By continuity of the eigenvalue in μ , this property persists for nearby μ , extending monotonicity to the tail. See Appendix I for details.

Closed. Let $\mu_n \rightarrow \mu^*$ with $\mu_n \in \mathcal{S}$. By continuity, $H_t(\mu_n) \rightarrow H_t(\mu^*)$ for each t . Since each $H_t(\mu_n)$ has constant nonzero sign, the limit inherits constant weak sign—say $H_t(\mu^*) \geq 0$ for all t —so the K -path is weakly monotone.

It remains to show strict inequality. By the eigenvalue argument from the Open step, $H_t(\mu^*) > 0$ for all sufficiently large t , so any zero of H must occur at a finite t^* with $K_{t^*}(\mu^*) \neq K_A^{cs}(\mu^*)$. $H_{t^*} = 0$ means $\Delta K_{t^*} = 0$: the point (K_{t^*}, C_{t^*}) lies on the distribution-free K -nullcline (Proposition 7) at a non-steady-state capital level. This contradicts *unique K -stationarity*: the property that $\Delta K_t = 0$ along the convergent path only at K_A^{cs} . In the representative-agent model, K is Markov under the frozen regime, so $\Delta K_{t^*} = 0$ makes K_{t^*} a fixed point; uniqueness of the conditional steady state (Assumption 1(i)) yields the contradiction directly. In the heterogeneous-agent model, the eigenvalue condition $\lambda_s(\mu^*) \in (0, 1)$ establishes unique K -stationarity locally near $K_A^{cs}(\mu^*)$; globally, unique K -stationarity is maintained as part of hypothesis (ii) of the proposition and verified pathwise by direct inspection of the simulated continuation. So $H_t(\mu^*) > 0$ for all t and $\mu^* \in \mathcal{S}$. ■

Strict K -monotonicity is equivalent to the endpoint no-side-switching condition $H_t H_{t+1} > 0$ for every pre-limit t , where $H_t := C_t - C_A^{Knull}(K_t)$ is the nullcline gap. Combined with the curvature-margin check of Lemma G.2 (Appendix G), the proposition delivers the full line-segment condition of Assumption 2. Condition (ii) is automatic in the representative-agent model (uniqueness of the conditional steady state, Assumption 1(i)); in the heterogeneous-agent model, it holds locally via the stable-eigenvalue condition (Proposition 10) and is maintained globally as a pathwise condition, directly verifiable along the simulated continuation. The bounded price-income MPC of Assumption 3 and the path-level FD bound of Assumption 4 (Appendix G) together rule out side-switching of the nullcline gap, but neither by itself rules out non-steady-state flat steps.

Discussion Proposition 11 reduces verification to a perfect-foresight monotonicity check, paired with the no-plateau and uniform-regularity conditions in hypotheses (ii) and (iii). The PF check is a textbook property of the Ramsey saddle in the representative-agent case; in the heterogeneous-agent case, monotone convergence of aggregate capital along computed Bewley–Aiyagari transitions under fixed TFP is a robust numerical regularity (Lee, 2025), though not a general theorem. The proof connects $\mu = 0$ to $\mu = 1$ through a continuous path of equilibria, using nullcline invariance (Proposition 7) as the common “obstacle.” The heterogeneous-agent case requires unique K -stationarity at every $\mu \in [0, 1]$ to rule out non-steady-state flat steps, and uniform spectral regularity (Proposition 10, Appendix H) to control the tail along the homotopy. These are maintained as hypotheses and admit direct numerical verification along the simulated continuation; the MPC-based primitives of Section 4.2 additionally rule out endpoint side-switching but do not by themselves discharge unique K -stationarity.

The diagnostic extends to any continuous aggregate variable e satisfying unique stationarity ($e(\Phi_{t+1}) = e(\Phi_t)$ only at $\Phi_t = \Phi^{\text{cs}}$), even when the nullcline is not distribution-free. This matters for the extensions in Section 4.3. The CRRA-separable labor case makes the nullcline distribution-dependent; at $\mu = 0$, however, the aggregate dynamics reduce to a single deterministic transition path, so monotonicity needs to be checked only along that path.

4.2 Primitive conditions for Assumption 2

An independent verification route works directly under rational expectations ($\mu = 1$). It uses static allocation primitives—bounds on household marginal propensities to consume and general-equilibrium price feedback—rather than solving the full stochastic equilibrium. The check is conditional on a realized frozen-regime continuation (or certified envelope bounds on the relevant moments along it). Verification proceeds in two steps (Appendix G). Lemma G.1 gives a primitive condition on MPCs and GE price feedback that rules out endpoint side-switching of the nullcline gap: $H_t H_{t+1} > 0$ for every t with $K_{t+1} \neq K_t$. Lemma G.2 then upgrades endpoint no-side-switching to the full line-segment no-crossing of Assumption 2 through a curvature-margin check: the nullcline gap at the endpoints must exceed $M(K_{t+1} - K_t)^2/8$, where M bounds the curvature of the nullcline on the traversed interval. Both steps admit direct numerical verification along the simulated continuation.

Appendix G presents the primitive MPC condition as a three-level hierarchy. Level 1 covers log utility with Cobb–Douglas production and bounded MPCs; it gives an explicit pointwise bound at the conditional steady state. Under the Krusell and Smith (1998) calibration, the inequality holds with substantial slack (LHS ≈ 0.04 , threshold 1). Level 2 extends Level 1 numerically to CRRA utility with σ in the empirically relevant range. Level 3 is the perfect-foresight diagnostic of Proposition 11, a numerical fallback when closed-form primitives cannot be verified. The hierarchy has real bite: Appendix G illustrates a non-concave production economy where the primitive condition cannot rule out segment crossing, confirming that Assumption 2 is a substantive restriction.

The hierarchy culminates in a theorem with two complementary parts. Near the conditional steady state, no MPC bound and no pathwise condition are needed: local spectral regularity of the equilibrium law of motion alone delivers exact K -sufficiency. Away from the steady state, the log–Cobb–Douglas structure converts the verification chain—no-side-switching, segment no-crossing, and the relevant moment bounds—into explicit inequalities. The following definition collects them.

Definition 7 (K -monotonicity interval).

Fix $A \in \{B, G\}$ and constants $\bar{m} \in (0, 1]$, $\bar{Z} > 0$, $\bar{a} > 0$. Define

$$\begin{aligned} B_A(K) &:= \bar{m} \alpha (1 - \alpha) A [K^{\alpha-1} \bar{Z} + K^{\alpha-2} \bar{a}], \\ N_A(K) &:= \alpha A K^{\alpha-1} - \delta, \\ M_A(K) &:= \alpha (1 - \alpha) A K^{\alpha-2}. \end{aligned}$$

An interval $[\underline{K}, \bar{K}]$ with $0 < \underline{K}$ and $\bar{K} \leq (A/\delta)^{1/(1-\alpha)}$ is a K -monotonicity interval with margin $\kappa \in (0, 1)$ if

$$B_A(\underline{K}) - N_A(\bar{K}) \leq 1 - \kappa \quad \text{and} \quad 2f(\bar{K}; A) + (1 - \delta)\bar{K} < \frac{8\kappa}{M_A(\underline{K})}.$$

In the theorem below, the constants are supplied by Assumptions 3–4 (Appendix G): $\bar{m}$ is the MPC bound, and $\bar{Z}$, $\bar{a}$ are the path bounds on mean productivity and mean assets. Each piece of the definition has a transparent role. B_A is the MPC–price-feedback term, N_A the nullcline slope, and M_A the nullcline curvature; since B_A , N_A , M_A are decreasing in K and $2f + (1 - \delta)K$ is increasing, the two endpoint inequalities bound the relevant extrema over the entire interval. The cap $\bar{K} \leq (A/\delta)^{1/(1-\alpha)}$ keeps the nullcline nonnegative. The first inequality is the no-side-switching margin; the second is the curvature margin that upgrades endpoint sign preservation to segment no-crossing.

Theorem 3 (Within-spell K -sufficiency under log utility and Cobb–Douglas).

Fix $A \in \{B, G\}$ and a conditional saddle $\mathcal{M}(\Phi_0, A)$ in the heterogeneous-household economy of this section, with log utility, Cobb–Douglas production, and compact idiosyncratic supports.

- (a) Suppose the regularity conditions of Proposition 10 hold at Φ_A^{cs} with stable eigenvalue $\lambda_A^s \in (0, 1)$. Then there exists a chart neighborhood $\mathcal{V}_A$ of Φ_A^{cs} such that, along the frozen- A continuation started from any $\Phi'_0 \in \mathcal{M}(\Phi_0, A) \cap \mathcal{V}_A$: (i) $\{K_t\}$ is strictly monotone and converges to K_A^{cs} without crossing it; (ii) $\Delta K_t = 0$ only at $\Phi_t = \Phi_A^{cs}$; and (iii) K is an exact conditional sufficient statistic on $\mathcal{M}(\Phi_0, A) \cap \mathcal{V}_A$.
- (b) Suppose, in addition, that Assumptions 3 and 4 hold, and let $\mathcal{K} = [\underline{K}, \bar{K}]$ be a K -monotonicity interval with margin $\kappa \in (0, 1)$. If (i) the K -path of the continuation, together with the segments connecting consecutive capital levels, remains in $\mathcal{K}$, and (ii) every K -stationary state of the continuation with capital in $\mathcal{K}$ belongs to $\mathcal{V}_A$, then Assumption 2 holds along the continuation, $\{K_t\}$ is strictly monotone until convergence, and K is an exact conditional sufficient statistic on the entire continuation.

Proof.

See Appendix G. ■

Numerical verification under the Krusell and Smith (1998) calibration The ergodic range $[\underline{K}, \overline{K}] = [34.6, 38.5]$ is a K -monotonicity interval with margin $\kappa = 0.95$: $B_A(\underline{K}) - N_A(\overline{K}) \approx 0.04 < 0.05$, and the curvature inequality holds with a margin exceeding two orders of magnitude. The productivity bound $\bar{Z} \leq z_{\max}$ is a priori from compact support; $\bar{a}$ and the ergodic range are read off the computed equilibrium, so the check validates the computed continuation rather than replacing computation. Replacing $\bar{a}$ by the a priori feasibility cap $(A/\delta)^{1/(1-\alpha)} \approx 319$ still yields $B_A(\underline{K}) - N_A(\overline{K}) \approx 0.23 < 1$, so the inequalities can still be verified from calibrated parameters alone.

Three clarifications make precise what the theorem does and does not claim. First, part (a)’s hypotheses are not primitives in the preferences-and-technology sense: they are local spectral regularity conditions on the equilibrium law of motion (a simple real stable eigenvalue with a C^1 invariant curve containing the orbit tail; Appendix F). What part (a) establishes is that *given* this standard regularity, the entire pathwise apparatus—MPC bounds, segment-crossing restrictions, plateau conditions—is unnecessary near the conditional steady state. Note also that proximity in part (a) is in the distributional chart: membership of K_0 in $K(\mathcal{V}_A)$ alone is not sufficient, since distinct distributions can share the same capital. Second, part (a) is a statement about the late phase of a spell: it applies once the continuation has entered the (unquantified) linearization neighborhood, while coverage of the entire ergodic range, including spell starts inherited from the other regime, is the role of part (b). Third, part (b)’s region is explicit but not assumption-free: Assumption 4 is a maintained path-level bound whose derivation from the policy-level Assumption 3 remains open (Appendix G), and hypothesis (b)(ii)—no plateau outside the local state-space neighborhood—is the single residual condition separating this result from a fully unconditional global theorem. It is stated as a hypothesis rather than buried in the verification chain, and Appendix G discusses why closing it appears to require new machinery for the basin of attraction of the conditional steady state.

4.3 Extensions of the dimension-reduction result

Theorem 1 provides verifiable conditions for dimension reduction: identify the K -nullcline and verify monotonicity, using either the perfect-foresight diagnostic (Proposition 11) or the primitive MPC conditions (Appendix G). This subsection applies these tools to several extensions.

Economies with multiple endogenous states Theorem 1 applies to an economy with the multivariate (distributional) endogenous state. As long as there is an aggregate equilibrium variable e that strictly monotonically converges to the conditional steady-state level,

then e is a conditional sufficient statistic on each conditional saddle path.

For example, consider the following model that extends the heterogeneous-agent model above by adding endogenous bond holding. The corresponding budget constraint is:

$$c + a' + q(\Phi, A)b' = a(1 + r(\Phi, A)) + b + w(\Phi, A)z \quad (33)$$

where b is bond holding and q is the bond price competitively determined by

$$\int b'(a, z; \Phi, A) d\Phi = 0. \quad (34)$$

The key observation is that Proposition 7 continues to hold: the conditional K -nullcline remains $C_A^{Knull}(K) = AK^\alpha - \delta K$, determined solely by the aggregate resource constraint. Since bonds are in zero net supply, their inclusion does not alter the sign-determining property of the nullcline. Thus, the monotonicity argument of Proposition 8 applies, and K remains a conditional sufficient statistic. Lee (2025) confirmed this prediction computationally: equilibrium allocations are strictly monotonically sorted along K in the globally solved model, consistent with Theorem 1.

Multiple predetermined aggregate states When the economy features multiple predetermined aggregate states—e.g., liquid and illiquid capital in a two-asset HANK model—the conditional saddle is generically higher-dimensional, and Theorem 1 requires a monotone *scalar* coordinate along this higher-dimensional object. Existence of such a coordinate is model-specific and cannot be guaranteed by the K -nullcline argument alone. The perfect-foresight diagnostic (Proposition 11) remains applicable provided one can identify a monotone aggregate variable along the deterministic transition path. Establishing general sufficient conditions for dimension reduction on multi-dimensional conditional saddles is an open question.

Models with endogenous labor supply With GHH preferences $u(c, l_H) = \frac{1}{1-\sigma}(c - \frac{\eta}{1+1/\chi} l_H^{1+1/\chi})^{1-\sigma}$, labor supply depends only on wages and productivity, eliminating wealth effects. The conditional K -nullcline depends on technology parameters and the stationary productivity distribution but not on the wealth distribution (Appendix J), so the monotonicity argument of Theorem 1 applies and K remains a conditional sufficient statistic along each conditional saddle path.

Under separable CRRA utility $u(c, l_H) = \frac{c^{1-\sigma}}{1-\sigma} - \frac{\eta}{1+1/\chi} l_H^{1+1/\chi}$, individual labor supply depends on consumption through the wealth effect, so the K -nullcline acquires distributional dependence. Strict monotonicity cannot be verified analytically in this case, and establishing

it formally under CRRA-separable preferences remains an open question. The perfect-foresight diagnostic of Proposition 11 remains applicable since at $\mu = 0$ the aggregate dynamics follow a single deterministic transition path, and computational experiments under standard calibrations confirm strictly monotone convergence along it (Lee, 2025). The derivations for both specifications are in Appendix J.

The boundary of dimension reduction The extensions above chart where the dimension-reduction logic carries over; the framework is equally informative about where it stops. In the lumpy-investment economy of Khan and Thomas (2008), nullcline invariance fails—wealth effects on labor supply and threshold-driven aggregate investment make the resource constraint distribution-dependent—so the analytical route to K -sufficiency is blocked, and the within-spell monotonicity prediction of Proposition 8 becomes a testable diagnostic rather than a theorem. Appendix M runs the test across three economies: the Khan and Thomas (2008) model in general equilibrium, its partial-equilibrium variant with prices fixed, and the Krusell and Smith (1998) benchmark. The diagnostic is clean, and scalar-state forecasting accurate, in both general-equilibrium economies—rationalizing the finding of Khan and Thomas (2008) that approximate aggregation survives lumpy investment. Removing the price feedback breaks the monotonicity and the accuracy together: dynamic forecast errors rise thirtyfold even as the one-step R^2 stays at 0.998, echoing Den Haan (2010), and spells flagged by the diagnostic carry the largest errors. Where the diagnostic is clean, scalar-state methods are on solid ground; where it degrades, the cross-sectional distribution carries independent forecasting power (Bachmann et al., 2013).

5 Search and matching

The conditional saddle framework applies to any recursive competitive equilibrium with a predetermined endogenous state, not only neoclassical growth models. This section applies it to a Diamond–Mortensen–Pissarides search-and-matching economy, where the predetermined state is unemployment. Conditional saddle paths organize the joint dynamics of unemployment and vacancies: the regime-specific vacancy loci slope upward, and the downward-sloping Beveridge-curve pattern in (u, v) data emerges as a loop traced by transitions across them. The slope-asymmetry characterization of Lemma 1 and Proposition 5 acquires a concrete labor-market interpretation: the surplus compression that the fundamental-surplus analysis of Ljungqvist and Sargent (2017) places at the heart of the Shimer puzzle manifests as differential saddle steepness. The baseline model produces conditional saddles whose slope asymmetry generates state dependence in vacancy creation; comparative statics on unemploy-

ment benefits reveal a non-monotone relationship with volatility, traceable to convergence of the regime-specific conditional steady states at high benefit levels. Benefit-policy extensions are collected in Appendix K.

The DMP environment Consider a discrete-time stochastic search-and-matching economy in the Diamond–Mortensen–Pissarides tradition.¹⁸ Aggregate productivity follows a Markov chain $z \in \{G, B, \dots\}$; for exposition I focus on the two extreme states $z \in \{G, B\}$. The aggregate state is $S = [u_{-1}, z]$, where unemployment $u_{-1} = 1 - n_{-1}$ is the sole predetermined endogenous state. Employment evolves according to

$$n = (1 - \lambda) n_{-1} + q(\theta) v, \quad (35)$$

where $\lambda \in (0, 1)$ is the exogenous separation rate, $\theta = v/u$ is labor market tightness, and $q(\theta) = m_0 \theta^{-\xi}$ is the vacancy-filling probability implied by a Cobb–Douglas matching function. Market tightness $\theta(S)$ —and hence vacancies $v(S) = \theta(S) \cdot u_{-1}$ —is the jump variable. It is pinned down by the free-entry condition for vacancy posting:

$$\frac{\kappa}{q(S)} = \beta \mathbb{E} \left[\left(\frac{c(S)}{c(S')} \right)^\sigma \left(z' - w(S') + (1 - \lambda) \frac{\kappa}{q(S')} \right) \middle| S \right], \quad (36)$$

where κ is the vacancy posting cost, $w(S)$ is the Nash-bargained wage, and $c(S) = n(S) z - \kappa v(S) + (1 - n(S)) b$ is aggregate consumption from the resource constraint.

The conditional saddle path $\mathcal{M}(u_0, z)$ is the orbit $\{(u_t, v_t)\}_{t \geq 0}$ generated by (35) and the equilibrium vacancy function $v(\cdot, z)$ under frozen productivity z . Its limit point is $(u_z^{\text{cs}}, v_z^{\text{cs}})$.

Slope asymmetry and the Beveridge curve Figure 10 plots the conditional saddle paths in (u_{-1}, v) space. Both loci slope upward—higher inherited unemployment raises equilibrium vacancies—so state dependence is sign-preserving, as in the RBC benchmark. However, the slopes differ markedly. Under G , matches are profitable and the saddle is steep; under B , the flow surplus is low and the saddle is nearly flat.

The mechanism is the free-entry condition (36). Each additional unemployed worker represents a potential match, but the value of that match depends on productivity. Under high z , the flow surplus $z - w$ is large, so firms respond aggressively to a marginal increase in the unemployment pool, producing a steep vacancy locus. Under low z , the surplus is

¹⁸The full model specification and calibration are in Appendix K. I adopt a standard formulation following Shimer (2005) with Nash-bargained wages, exogenous separation, and Cobb–Douglas matching. The computation uses an 11-state Tauchen discretization of log TFP; for expositional simplicity, I present the two extreme states as G and B .

compressed—with the outside option b fixed, the Nash wage falls less than one-for-one with productivity, squeezing $z - w$ (Hagedorn and Manovskii, 2008; Ljungqvist and Sargent, 2017)—so vacancy posting barely responds, producing a nearly flat locus. The slope asymmetry is the visual signature of the surplus compression that underlies the unemployment-volatility puzzle of Shimer (2005).

This asymmetry generates the Beveridge curve “loop” observed in empirical (u, v) data. Starting near the G steady state, a negative shock switches the economy onto the flat B saddle. Vacancies collapse on impact and barely adjust as unemployment accumulates along the flat path. Upon recovery, the economy jumps back onto the steep G saddle at high inherited u_{-1} , driving a sharp vacancy spike and rapid job creation. The counterclockwise rotation through recessions and recoveries is thus the across-saddle jump dynamics of the stochastic equilibrium.

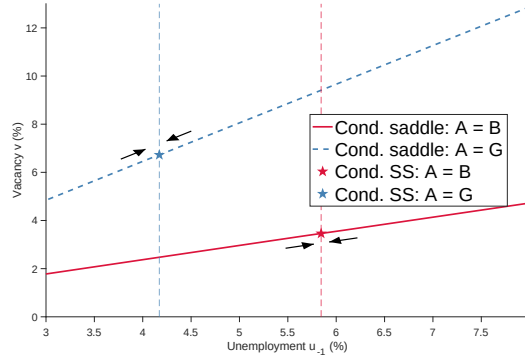


Figure 10: Conditional saddle paths in the DMP model, constant b

Notes: Conditional saddle paths in (u_{-1}, v) space under frozen productivity $z \in \{G, B\}$. Stars mark conditional steady states; arrows indicate the direction of convergence. Extensions with stochastic and unemployment-dependent benefits are in Appendix K.

Comparative statics of the benefit level Ergodic unemployment volatility is hump-shaped in the unemployment-benefit level b , with an interior peak. The operative geometric diagnostic is the separation between regime-specific conditional steady states, $u_B^{\text{cs}}(b) - u_G^{\text{cs}}(b)$, not slope asymmetry alone. At high b , u_B^{cs} saturates while u_G^{cs} keeps rising; the steady-state gap collapses, and the synchronization contraction (Theorem 2(i), with the uniform-contraction hypothesis maintained for the scalar-state DMP frozen-regime map, where it is directly checkable) anchors each regime’s ergodic states near its conditional steady state, so the collapsing gap compresses the range over which unemployment can fluctuate between regimes. The mechanism is the asymmetric surplus compression implicit in the Nash wage—the same fundamental-surplus channel through which Ljungqvist and Sargent (2017) show

the benefit level governs amplification across DMP variants; the saddle geometry adds the steady-state-separation diagnostic that makes the non-monotonicity visible. Robustness to wage-setting rules and matching elasticities is documented in Appendix K.

How does the benefit level b reshape the conditional saddle geometry, and what does this imply for unemployment volatility? The Nash wage $w = (1 - \eta)b + \eta(z + \kappa\theta)$ is increasing in b , so raising b compresses the match surplus $z - w$. The compression is proportionally more severe in the B regime, where $z - b$ is already small. Figure 11a solves the DMP model for $b \in \{0.10, 0.40, 0.70, 0.90\}$. At every benefit level, the G saddle is steeper than the B saddle; as b rises, the deepening surplus compression flattens both loci—leaving the B saddle nearly flat—while pushing the conditional steady states toward higher unemployment. One might expect the deepening compression to raise volatility monotonically, but the effect is *hump-shaped* (Figure 11b): volatility peaks at $b \approx 0.55$ ($\text{std}(\log u) \approx 7.6\%$, with a nearly flat top through $b \approx 0.60$) and then *declines*. The exact peak location is model- and calibration-specific. Two clarifications guard against misreading. First, the result is a *ceteris-paribus* comparative static: the amplification in Hagedorn and Manovskii (2008) at $b \approx 0.955$ operates through their joint recalibration of the bargaining weight and the vacancy-posting cost, not through the benefit level alone, so the hump here neither contradicts nor overturns their calibration. Second, holding all other parameters fixed, the saddle geometry explains why b alone is hump-shaped: at high b , conditional steady states converge despite large slope gaps, compressing the range over which unemployment can fluctuate.¹⁹

The saddle framework diagnoses this hump by revealing that what governs volatility is the *separation between conditional steady states*, not slope asymmetry alone. The right subplot of Figure 11b plots conditional steady-state unemployment under frozen good (u_G^{cs}) and bad (u_B^{cs}) TFP for each b . As b rises, u_G^{cs} increases steadily while u_B^{cs} rises faster, widening the gap from under 1 p.p. at $b = 0.05$ to about 2.2 p.p. at $b \approx 0.60$. Beyond that point, u_B^{cs} *saturates*. Agents expect future TFP recovery, so vacancy posting persists even under bad productivity, capping the bad-state conditional steady state near 6.8%. Meanwhile, u_G^{cs} continues to rise as higher b compresses the surplus even under good TFP. The two conditional steady states therefore *converge* at high b : the saddles remain differentially tilted but sit close together, leaving little room for unemployment to fluctuate between regimes. Both the gap $u_B^{\text{cs}} - u_G^{\text{cs}}$ and simulated volatility are hump-shaped in b , peaking near the same value. The conditional steady-state separation—not the slope difference—is the operative geometric diagnostic for unemployment volatility.

¹⁹In the parameterization here, simulated volatility at $b = 0.955$ is about 4.2%, compared with 6.8% at the baseline $b = 0.40$. The saddle diagnostic identifies steady-state convergence as the mechanism.

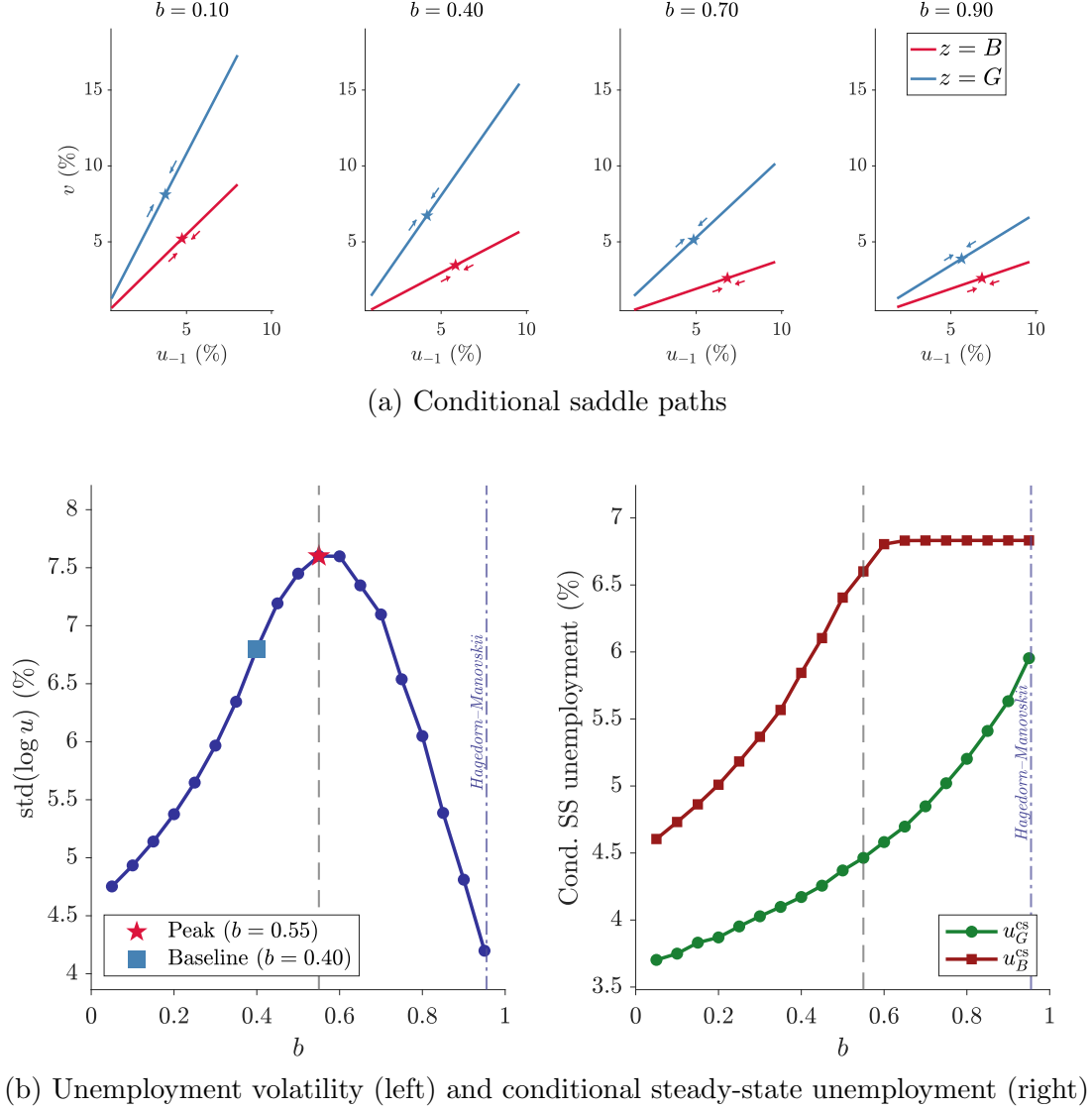


Figure 11: Comparative statics of b

Notes: Panel (a): conditional saddle paths in (u_{-1}, v) space for $b \in \{0.10, 0.40, 0.70, 0.90\}$; stars mark conditional steady states. Panel (b), left: $\text{std}(\log u)$ over 19 values of b ; the star marks the peak ($b \approx 0.55$), the square marks the baseline ($b = 0.40$), and the dash-dotted vertical line marks the [Hagedorn and Manovskii \(2008\)](#) replacement rate ($b = 0.955$). Panel (b), right: conditional steady-state unemployment under frozen good (u_G^{cs} , green) and bad (u_B^{cs} , red) TFP, extracted from the stochastic policy functions. Both the gap $u_B^{\text{cs}} - u_G^{\text{cs}}$ and simulated volatility are hump-shaped in b .

Sensitivity to wage setting and matching elasticity The hump-shaped relationship between b and unemployment volatility is robust across wage-setting rules and for matching elasticities up to the baseline value (Figure K.2). Under Nash bargaining with matching elasticities $\xi \in \{0.40, 0.50, 0.635, 0.80\}$, the hump persists for $\xi \leq 0.635$, with the peak shifting leftward as ξ increases: from $b \approx 0.80$ at $\xi = 0.40$ to $b \approx 0.55$ at the baseline $\xi = 0.635$. Higher matching elasticity amplifies the vacancy response to productivity differences, so

surplus compression sets in at lower b . At $\xi = 0.80$, volatility is nearly flat in b —the matching function is sufficiently elastic that conditional steady states converge over the entire range. Under *present-wage* bargaining, $w = (1 - \eta)b + \eta z$, which eliminates the congestion externality feedback from tightness to wages, the hump shape persists with peak volatility at $b \approx 0.93$. The non-monotone pattern is driven by surplus geometry rather than the specific wage-feedback channel.

6 Concluding remarks

Conditional saddle paths—equilibrium trajectories under frozen exogenous regimes—bring the geometric clarity of deterministic phase diagrams to stochastic environments. The slopes of these objects encode state dependence: differential tilt across saddles is the geometric signature of asymmetric shock responses. Their monotonicity delivers dimension reduction: a strictly monotone aggregate variable exactly indexes the equilibrium state within each frozen-regime spell. Persistent spells, exact within-spell sufficiency, and nullcline invariance constraining across-spell variation explain why this conditional result translates into near-global accuracy, providing a structural foundation for scalar-state methods.

The paper illustrates the framework through three applications. The canonical RBC model establishes the basic geometry and reveals that state-dependent response shapes arise from distance along the saddle even when slopes are nearly equal. The heterogeneous-household model of [Krusell and Smith \(1998\)](#) demonstrates that strict monotonicity along conditional saddles delivers exact within-spell dimension reduction—locally under standard regularity, and globally on an explicit region whose conditions can be checked from calibrated parameters, with one residual assumption stated rather than hidden; the across-spell error is bounded by how quickly the economy forgets its regime history. A non-concave production illustration shows the no-crossing condition is a genuine geometric restriction, and a lumpy-investment experiment confirms the diagnostic tracks where scalar-state accuracy degrades. The search-and-matching application shows the framework’s reach beyond neoclassical growth: the Beveridge-curve loop emerges from transitions across regime-specific saddle loci, surplus compression under low productivity manifests as differential saddle steepness, and the saddle geometry provides a direct diagnostic for the non-monotone relationship between benefit generosity and unemployment volatility.

More broadly, conditional saddles offer a complementary lens on stochastic models. They provide a language for interpreting nonlinear dynamics, for assessing when scalar state reductions are exact on the equilibrium path rather than merely accurate, and for characterizing across-path variation as a structural feature of the geometry. A promising direction for future

work is to use these geometric objects to sharpen empirical restrictions on state dependence and to discipline the design of state-contingent stabilization policies.

References

- ACHDOU, Y., J. HAN, J.-M. LASRY, P.-L. LIONS, AND B. MOLL (2021): “Income and Wealth Distribution in Macroeconomics: A Continuous-Time Approach,” *The Review of Economic Studies*, 89, 45–86.
- AIYAGARI, S. R. (1994): “Uninsured Idiosyncratic Risk and Aggregate Saving,” *The Quarterly Journal of Economics*, 109, 659–684, publisher: Oxford University Press.
- AKERLOF, G. A. AND J. L. YELLEN (1985): “Can Small Deviations from Rationality Make Significant Differences to Economic Equilibria?” *American Economic Review*, 75, 708–720.
- ANDREASEN, M. M., J. FERNÁNDEZ-VILLAYERDE, AND J. F. RUBIO-RAMÍREZ (2017): “The Pruned State-Space System for Non-Linear DSGE Models: Theory and Empirical Applications,” *The Review of Economic Studies*, 85, 1–49.
- ARNOLD, L. (1998): *Random Dynamical Systems*, Springer Monographs in Mathematics, Berlin, Heidelberg: Springer Berlin Heidelberg.
- AUCLERT, A., B. BARDÓCZY, M. ROGNLIE, AND L. STRAUB (2021): “Using the Sequence-Space Jacobian to Solve and Estimate Heterogeneous-Agent Models,” *Econometrica*, 89, 2375–2408.
- AUERBACH, A. J. AND Y. GORODNICHENKO (2012): “Measuring the Output Responses to Fiscal Policy,” *American Economic Journal: Economic Policy*, 4, 1–27.
- BACHMANN, R., R. J. CABALLERO, AND E. M. R. A. ENGEL (2013): “Aggregate Implications of Lumpy Investment: New Evidence and a DSGE Model,” *American Economic Journal: Macroeconomics*, 5, 29–67.
- BALEY, I. AND A. BLANCO (2019): “Firm Uncertainty Cycles and the Propagation of Nominal Shocks,” *American Economic Journal: Macroeconomics*, 11, 276–337.
- BARRO, R. J. AND X. SALA-I MARTIN (1992): “Convergence,” *Journal of Political Economy*, 100, 223–251.
- BASU, S. AND B. BUNDICK (2017): “Uncertainty Shocks in a Model of Effective Demand,” *Econometrica*, 85, 937–958.
- BERGER, D., K. MILBRADT, F. TOURRE, AND J. VAVRA (2021): “Mortgage Prepayment and Path-Dependent Effects of Monetary Policy,” *American Economic Review*, 111, 2829–78.

- BERGER, D. AND J. VAVRA (2015): “Consumption Dynamics During Recessions: Consumption Dynamics During Recessions,” *Econometrica*, 83, 101–154.
- BEWLEY, T. (1986): “Stationary Monetary Equilibrium with a Continuum of Independently Fluctuating Consumers,” *Contributions to Mathematical Economics in Honor of Gérard Debreu*, 79–102.
- BILAL, A. (2023): “Solving Heterogeneous Agent Models with the Master Equation,” Working Paper 31103, National Bureau of Economic Research.
- BLANCHARD, O. J. AND C. M. KAHN (1980): “The Solution of Linear Difference Models under Rational Expectations,” *Econometrica*, 48, 1305–1311.
- BLOOM, N., M. FLOETOTTO, N. JAIMOVICH, I. SAPORTA-EKSTEN, AND S. J. TERRY (2018): “Really Uncertain Business Cycles,” *Econometrica*, 86, 1031–1065.
- BOPPART, T., P. KRUSELL, AND K. MITMAN (2018): “Exploiting MIT shocks in heterogeneous-agent economies: the impulse response as a numerical derivative,” *Journal of Economic Dynamics and Control*, 89, 68–92.
- BOURANY, T. (2018): “Wealth Distribution over the Business Cycle: A Mean-Field Game with Common Noise,” Master’s thesis, UPMC-Sorbonne Université, supervised by Yves Achdou, Paris-Diderot University (LJLL).
- CARROLL, C. D. (1997): “Buffer-Stock Saving and the Life Cycle/Permanent Income Hypothesis,” *The Quarterly Journal of Economics*, 112, 56.
- CASS, D. (1965): “Optimum Growth in an Aggregative Model of Capital Accumulation¹,” *The Review of Economic Studies*, 32, 233–240.
- DEATON, A. (1991): “Saving and Liquidity Constraints,” *Econometrica*, 59, 1221–1248.
- DECHERT, W. D. AND K. NISHIMURA (1983): “A Complete Characterization of Optimal Growth Paths in an Aggregated Model with a Non-Concave Production Function,” *Journal of Economic Theory*, 31, 332–354.
- DEN HAAN, W. J. (1996): “Heterogeneity, Aggregate Uncertainty, and the Short-Term Interest Rate,” *Journal of Business & Economic Statistics*, 14, 399–411.
- (2010): “Assessing the accuracy of the aggregate law of motion in models with heterogeneous agents,” *Journal of Economic Dynamics and Control*, 34, 79–99.
- DEN HAAN, W. J. AND A. MARCET (1990): “Solving the Stochastic Growth Model by Parameterizing Expectations,” *Journal of Business & Economic Statistics*, 8, 31–34.
- FERNÁNDEZ-VILLAYERDE, J., S. HURTADO, AND G. NUÑO (2023): “Financial Frictions and the Wealth Distribution,” *Econometrica*, 91, 869–901.
- GALI, J. (2008): *Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework and Its Applications*, Princeton University Press.

- GORMAN, W. M. (1961): “On a Class of Preference Fields,” *Metroeconomica*, 13, 53–56.
- HAGEDORN, M. AND I. MANOVSKII (2008): “The Cyclical Behavior of Equilibrium Unemployment and Vacancies Revisited,” *American Economic Review*, 98, 1692–1706.
- HUGGETT, M. (1993): “The Risk-Free Rate in Heterogeneous-Agent Incomplete-Insurance Economies,” *Journal of Economic Dynamics and Control*, 17, 953–969.
- JO, Y. J. AND S. ZUBAIRY (2025): “State-Dependent Government Spending Multipliers: Downward Nominal Wage Rigidity and Sources of Business Cycle Fluctuations,” *American Economic Journal: Macroeconomics*, 17, 379–413.
- KAMIHIGASHI, T. (2003): “Necessity of transversality conditions for stochastic problems,” *Journal of Economic Theory*, 109, 140–149.
- (2005): “Necessity of the transversality condition for stochastic models with bounded or CRRA utility,” *Journal of Economic Dynamics and Control*, 29, 1313–1329.
- KAPLAN, G., B. MOLL, AND G. L. VIOLANTE (2018): “Monetary Policy According to HANK,” *American Economic Review*, 108, 697–743.
- KAPLAN, G. AND G. L. VIOLANTE (2014): “A Model of the Consumption Response to Fiscal Stimulus Payments,” *Econometrica*, 82, 1199–1239.
- KHAN, A. AND J. K. THOMAS (2008): “Idiosyncratic Shocks and the Role of Non-convexities in Plant and Aggregate Investment Dynamics,” *Econometrica*, 76, 395–436.
- KOOP, G., M. PESARAN, AND S. M. POTTER (1996): “Impulse response analysis in nonlinear multivariate models,” *Journal of Econometrics*, 74, 119–147.
- KOOPMANS, T. (1963): “On the Concept of Optimal Economic Growth,” Cowles Foundation Discussion Papers 163, Cowles Foundation for Research in Economics, Yale University.
- KRUSELL, P. AND A. A. SMITH, JR. (1998): “Income and Wealth Heterogeneity in the Macroeconomy,” *Journal of Political Economy*, 106, 867–896.
- LEE, H. (2025): “Global Nonlinear Solutions in Sequence Space and the Generalized Transition Function,” Working Paper.
- (2026): “Striking While the Iron Is Cold: Fragility after a Surge of Lumpy Investments,” *Working Paper*, 68.
- LJUNGQVIST, L. AND T. J. SARGENT (2017): “The Fundamental Surplus,” *American Economic Review*, 107, 2630–2665.
- MARCET, A. (1988): “Solving nonlinear stochastic models by parameterizing expectations,” Manuscript. Pittsburgh: Carnegie Mellon Univ.
- MELCANGI, D. (2024): “Firms’ precautionary savings and employment during a credit crisis,” *American Economic Journal: Macroeconomics*, 16, 356–386.

- MENZIO, G. AND S. SHI (2010): “Block Recursive Equilibria for Stochastic Models of Search on the Job,” *Journal of Economic Theory*, 145, 1453–1494.
- (2011): “Efficient Search on the Job and the Business Cycle,” *Journal of Political Economy*, 119, 468–510.
- NEFTCI, S. N. (1984): “Are Economic Time Series Asymmetric over the Business Cycle?” *Journal of Political Economy*, 92, 307–328.
- PETROSKY-NADEAU, N., L. ZHANG, AND L.-A. KUEHN (2018): “Endogenous Disasters,” *American Economic Review*, 108, 2212–45.
- PIZZINELLI, C., K. THEODORIDIS, AND F. ZANETTI (2020): “STATE DEPENDENCE IN LABOR MARKET FLUCTUATIONS,” *International Economic Review*, 61, 1027–1072.
- PROEHL, E. (2025): “Existence and Uniqueness of Recursive Equilibria with Aggregate and Idiosyncratic Risk,” Working Paper.
- RAMSEY, F. P. (1928): “A Mathematical Theory of Saving,” *The Economic Journal*, 38, 543–559.
- REDDY, P. B., J. M. SCHUMACHER, AND J. C. ENGWERDA (2020): “A Bendixson Criterion for Optimal Control of a Class of Hybrid Systems,” *SIAM Journal on Control and Optimization*, 58, 2219–2252.
- REITER, M. (2010): “Solving the incomplete markets model with aggregate uncertainty by backward induction,” *Journal of Economic Dynamics and Control*, 34, 28–35, computational Suite of Models with Heterogeneous Agents: Incomplete Markets and Aggregate Uncertainty.
- SCHAAL, E. (2017): “Uncertainty and Unemployment,” *Econometrica*, 85, 1675–1721.
- SCHENK-HOPPÉ, K. R. (2001): “Random Dynamical Systems in Economics,” *Stochastics and Dynamics*.
- SCHENK-HOPPÉ, K. R. AND B. SCHMALFUSS (2001): “Random fixed points in a stochastic Solow growth model,” *Journal of Mathematical Economics*, 36, 19–30.
- SCHENK-HOPPÉ, K. R. (1998): “Random attractors—general properties, existence and applications to stochastic bifurcation theory,” *Discrete and Continuous Dynamical Systems*, 4, 99–130.
- SHIMER, R. (2005): “The Cyclical Behavior of Equilibrium Unemployment and Vacancies,” *American Economic Review*, 95, 25–49.
- SKIBA, A. K. (1978): “Optimal Growth with a Convex–Concave Production Function,” *Econometrica*, 46, 527–539.

- SOLOW, R. M. (1956): “A Contribution to the Theory of Economic Growth,” *The Quarterly Journal of Economics*, 70, 65–94.
- SWAN, T. W. (1956): “ECONOMIC GROWTH and CAPITAL ACCUMULATION,” *Economic Record*, 32, 334–361.
- TENREYRO, S. AND G. THWAITES (2016): “Pushing on a String: US Monetary Policy Is Less Powerful in Recessions,” *American Economic Journal: Macroeconomics*, 8, 43–74.
- VAVRA, J. (2014): “INFLATION DYNAMICS AND TIME-VARYING VOLATILITY: NEW EVIDENCE AND AN SS INTERPRETATION,” *The Quarterly Journal of Economics*, 129, 215–258.
- WAGENER, F. (2003): “Skiba Points and Heteroclinic Bifurcations, with Applications to the Shallow Lake System,” *Journal of Economic Dynamics and Control*, 27, 1533–1561.
- WALSH, K. J. AND E. YOUNG (2024): “Equilibrium Multiplicity in Aiyagari and Krusell-Smith,” Working paper.
- WINBERRY, T. (2021): “Lumpy Investment, Business Cycles, and Stimulus Policy,” *American Economic Review*, 111, 364–396.
- YANNAKOPOULOS, A. N. (2011): “Stochastic Saddle Paths and Economic Theory,” in *Dynamics, Games and Science II*, ed. by M. M. Peixoto, A. A. Pinto, and D. A. Rand, Berlin, Heidelberg: Springer Berlin Heidelberg, 735–752.